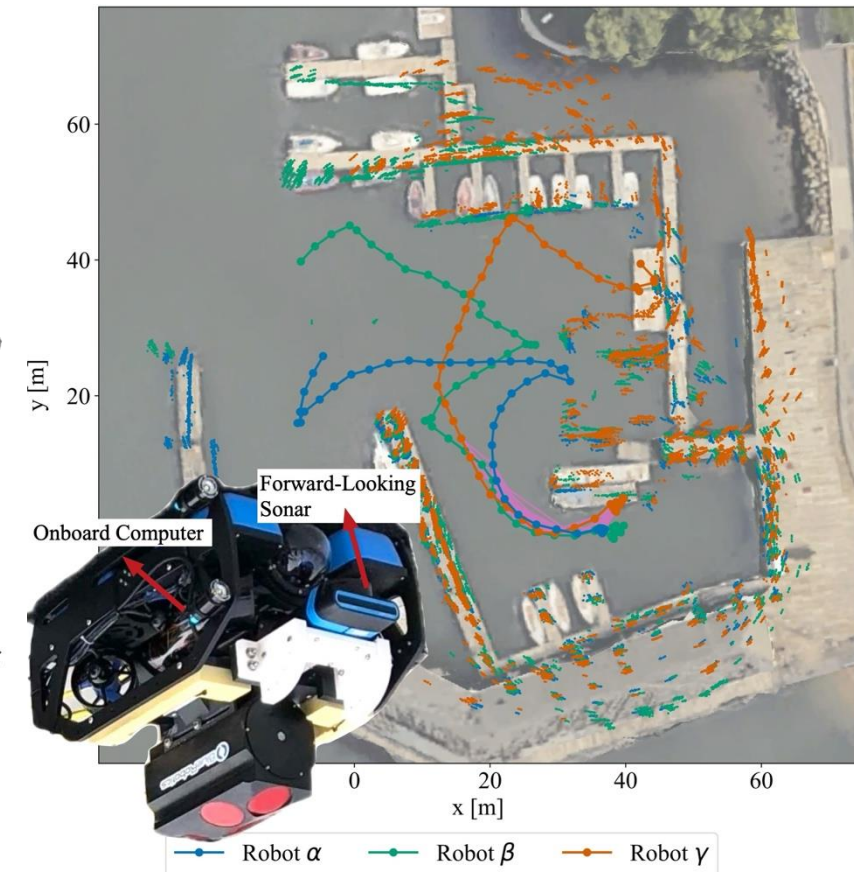
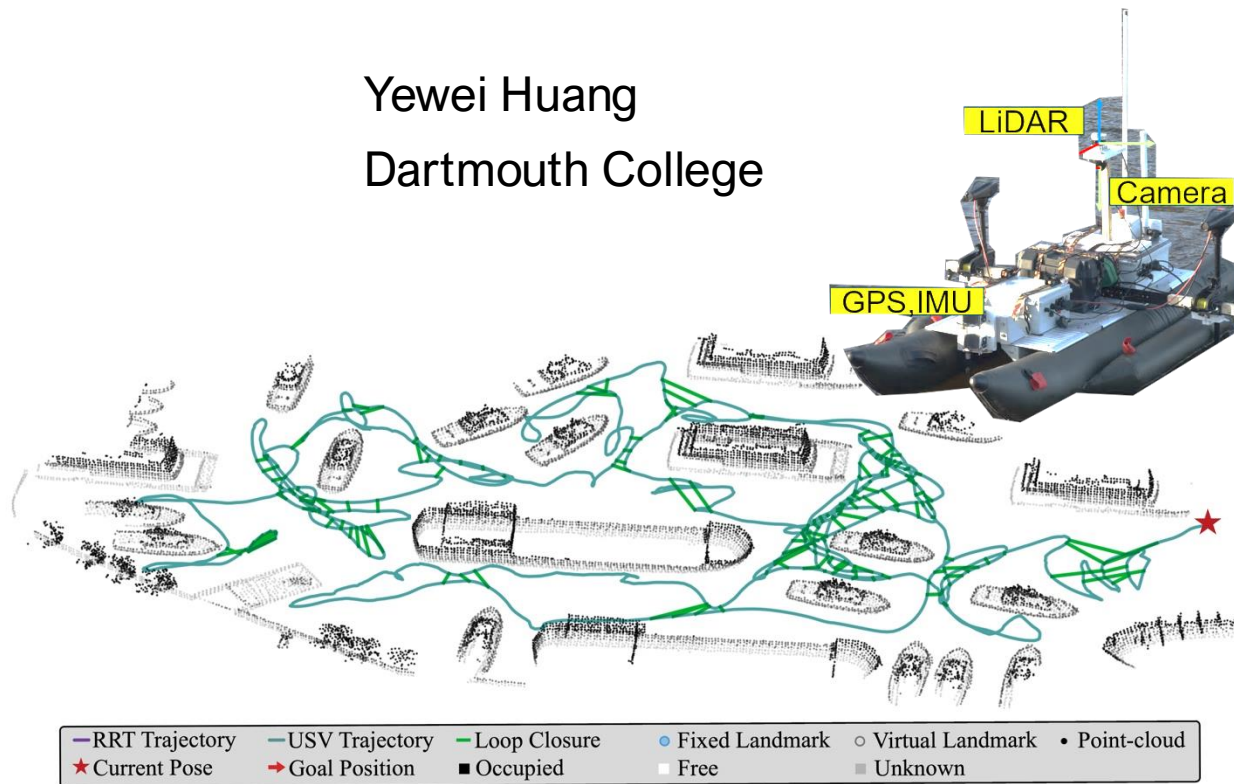


Autonomous Navigation and Mapping in Coastal Waters with Environmental Variability

Yewei Huang
Dartmouth College



Yewei Huang

Background

- Postdoctoral researcher in the [Reality and Robotics Lab](#) at the Department of Computer Science, Dartmouth College, advised by [Alberto Quattrini Li](#), Associate Professor.
- Ph.D. in Mechanical Engineering, Stevens Institute of Technology, advised by [Brendan Englot](#), Anson Wood Burchard Endowed Professor and Director of the Stevens Institute for Artificial Intelligence (SIAI).

Research Interest

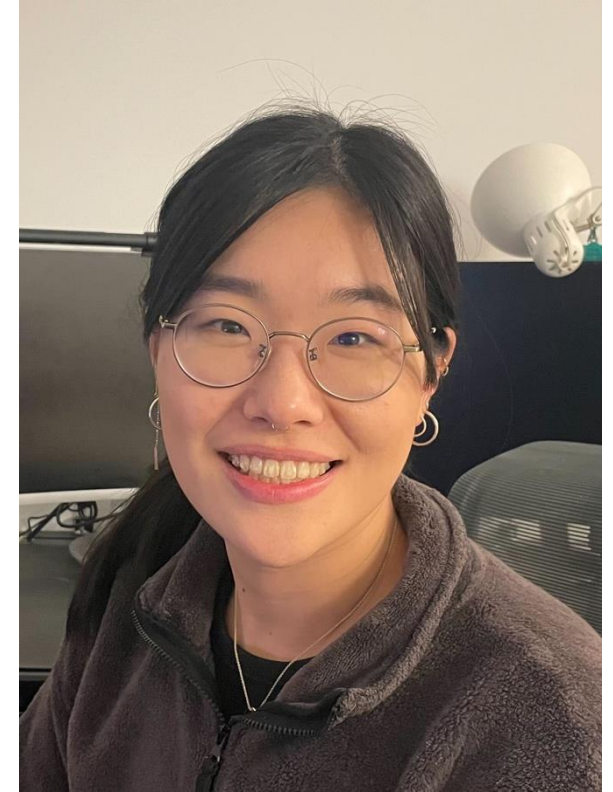
- Navigation and perception of the multi-robot teams, especially in the marine environment.

Check my latest work at: <https://yewei Huang.github.io/>

Check latest work of RFAL at:

<https://robustfieldautonomylab.github.io/>

Check latest work of R-Lab at: <https://rlab.cs.dartmouth.edu/home/>



Our Lab

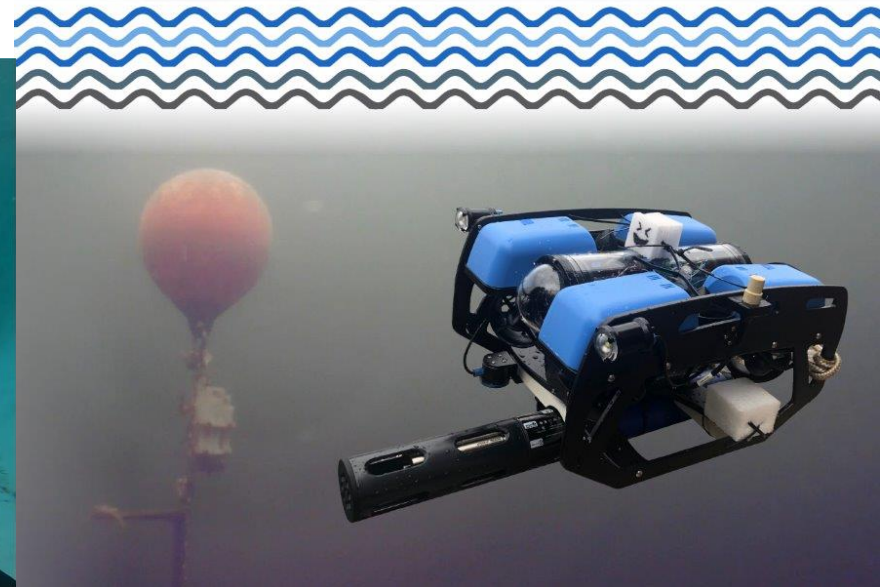
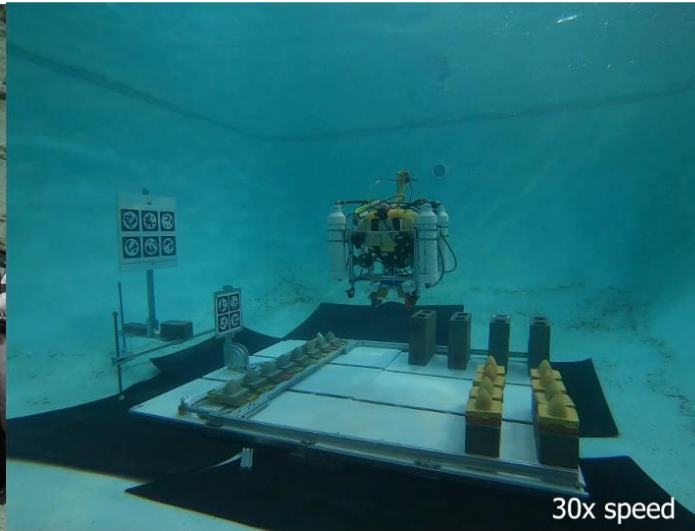
Research Interest

- Navigation and mapping of aqua environment
- Marine environmental monitoring
- Multi-robot collaboration in aqua environment

Always hiring and happy to collaborate!



DARTMOUTH



Why the robots need to navigate themselves in the coastal environment?

- Ports and harbor security



- Environmental monitoring



- Offshore energy infrastructure



- Disaster response



Why the robots need to navigate themselves in the coastal environment? -- Fundamental for all missions

- Ports and harbor security



- Environmental monitoring



- Offshore energy infrastructure



- Disaster response



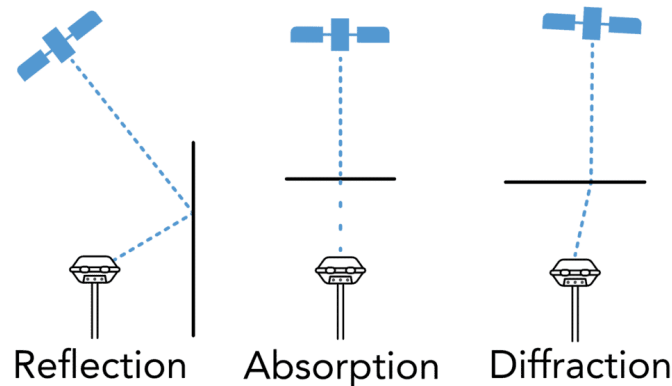
Why the reliable navigation and mapping remains a problem?

- **Autonomous Car (Waymo)**
 - High-precision GPS + HD maps
 - Mostly planar motion
 - Vision and LiDAR fusion
 - Rich, dense semantic structures
- **Autonomous Surface Vehicles (ASV)**
 - GPS available but **degraded**
 - Drift from **currents** and **wind**
 - Vision and LiDAR fusion
 - **Uneven** structures
- **Underwater Unmanned Vehicles (UUV)**
 - **No** GPS
 - Influenced by **hydro-dynamics**
 - Only sonar in murky water
 - **Unstructured**



How to navigate efficiently and effectively under restrictions?

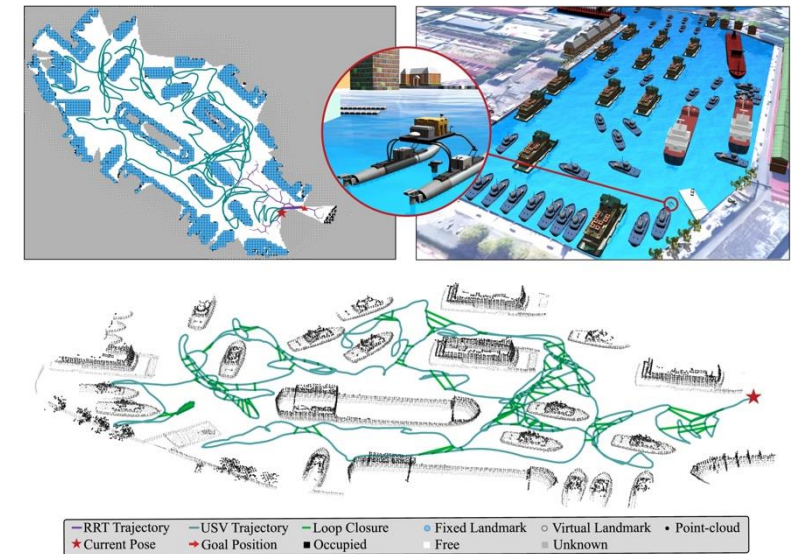
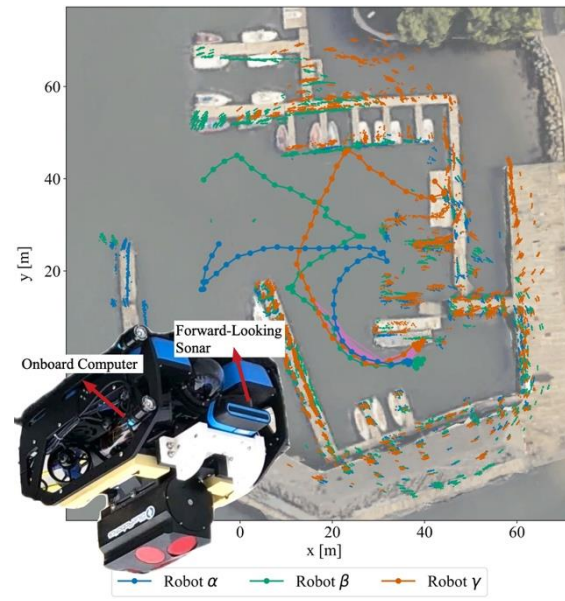
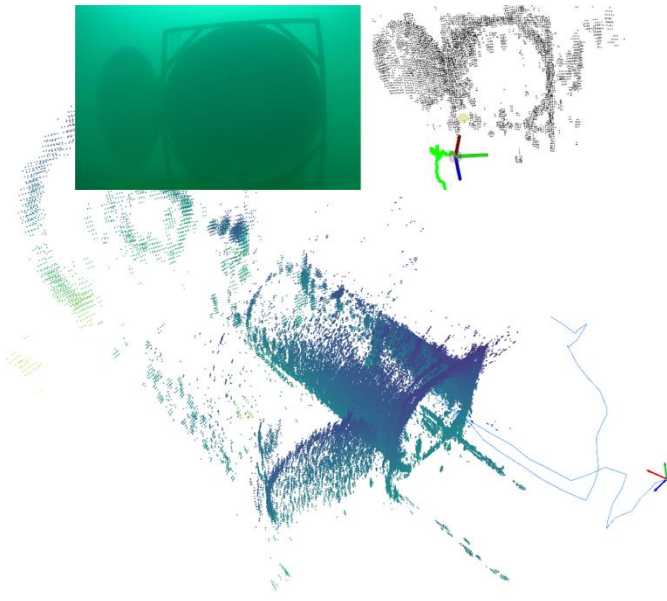
- **Energy constraints:** Mission duration is limited to several hours by onboard battery capacity
- **GPS-denied operation:** GPS is unreliable for coastal ASVs and unavailable for UUVs; vehicles must periodically surface to obtain position fixes
- **Underwater communication limits:** UUVs rely on acoustic links with ~65 kbps bandwidth (vs. Wi-Fi at 10–1000 Mbps), often with no communication until mission completion



How to navigate efficiently and effectively under restrictions?

We try to solve this problem from different perspectives:

- 3D sonar-based Simultaneous Localization and Mapping (SLAM) for UUV
- Multi-robot SLAM with multi-beam sonar under communication constraints for UUV
- Autonomous exploration using a variable-resolution virtual map for ASV



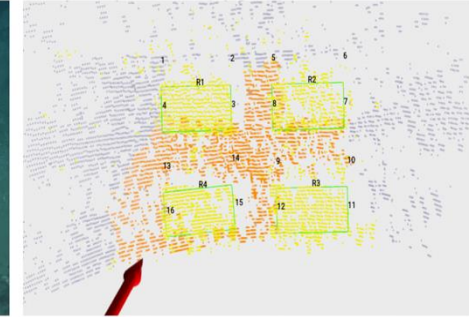
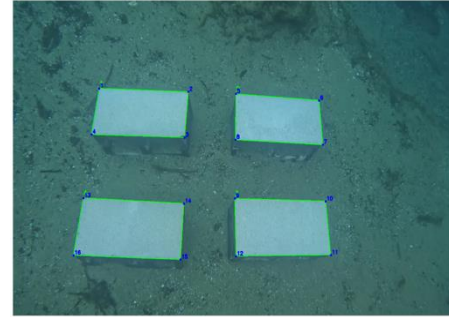
Differences between a multi-beam and a 3D sonar

- Multi-beam (fan-beam) sonar
 - 2D fan
 - Sparser Data



A buoy equipped with hydrophone

- Compact Volumetric 3D sonar
 - 3D volume
 - Denser Data



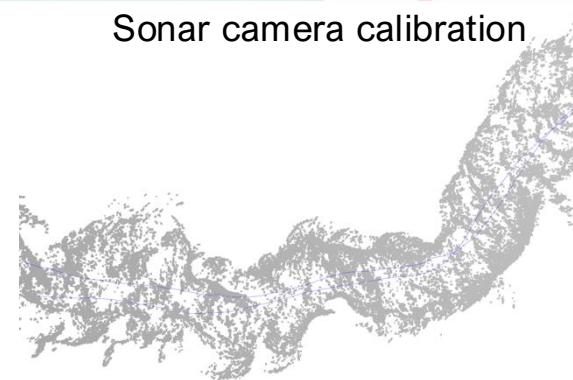
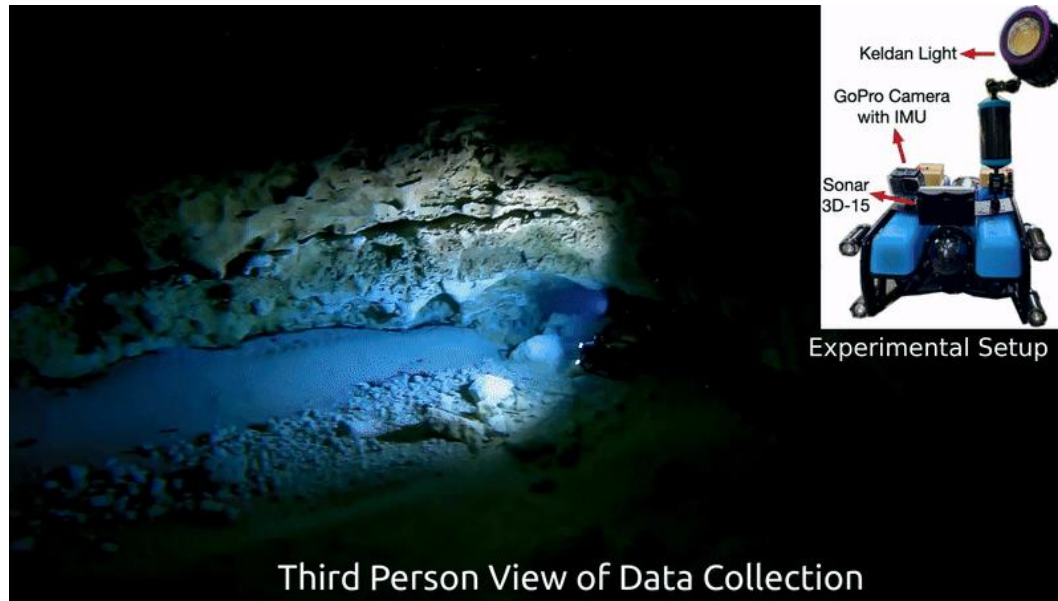
Volumetric 3D sonar scan



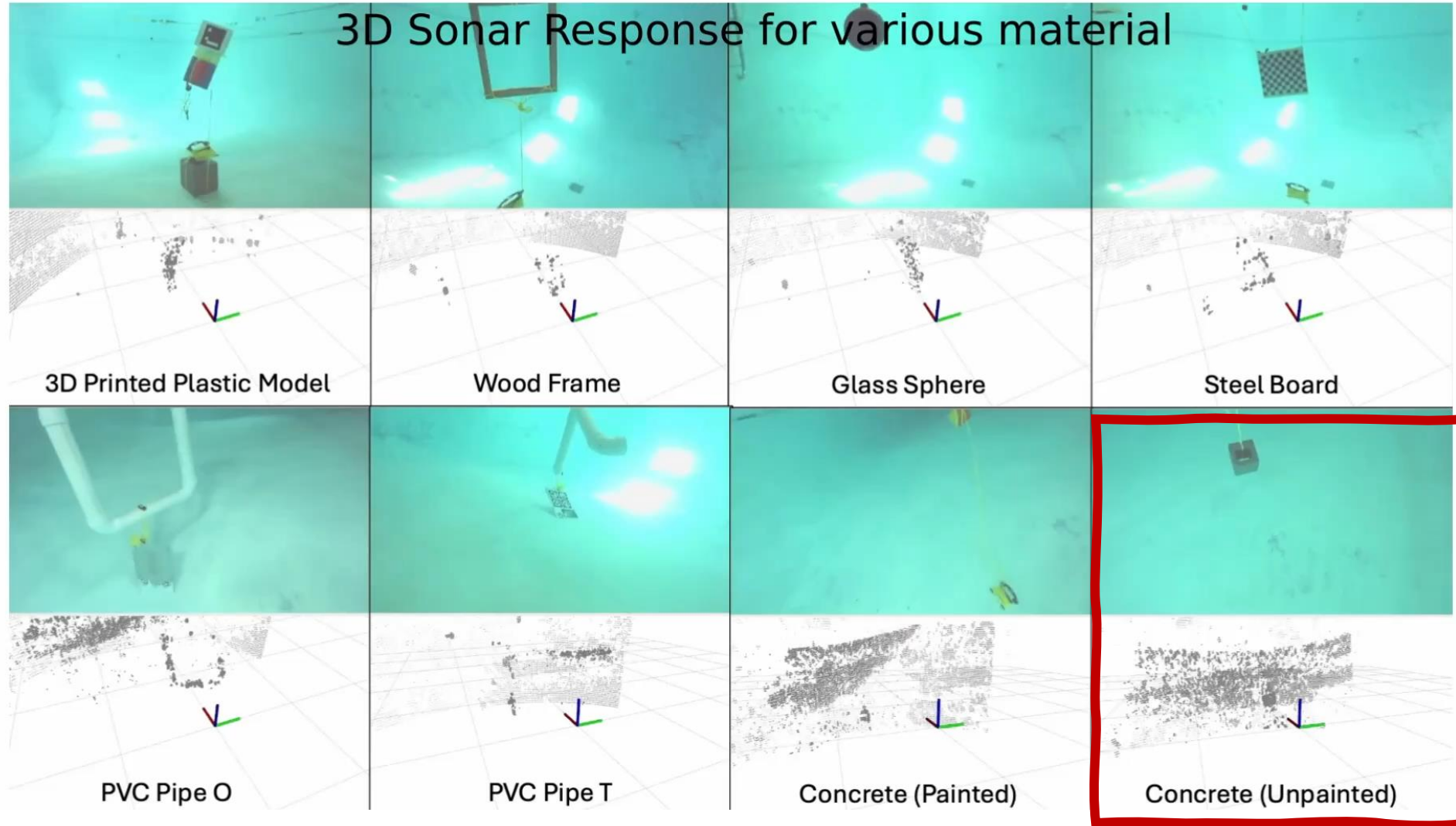
LiDAR reconstruction of a water dam

3D sonar-based Simultaneous Localization and Mapping (SLAM) for UUV

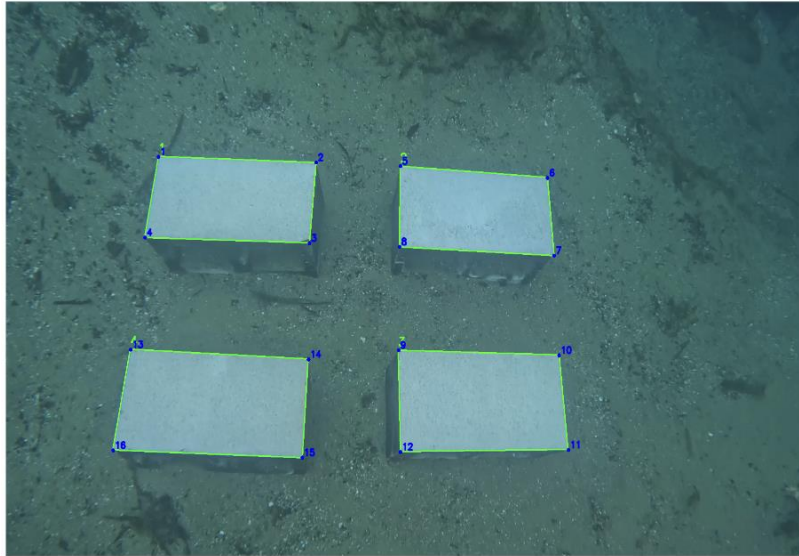
- Process pipeline : data gathering in the field + offline data processing



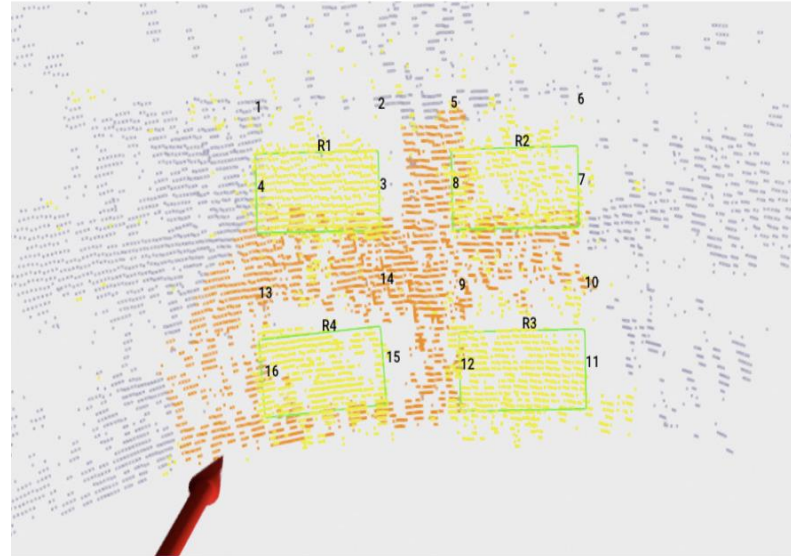
Camera Sonar Calibration



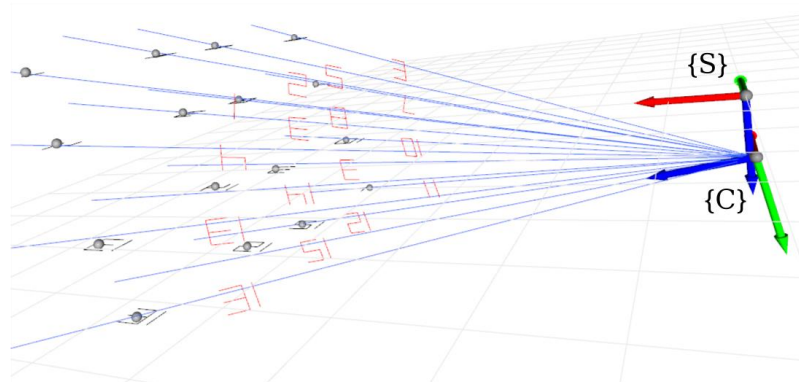
Camera Sonar Calibration



Extract 2D pixel position of 16 corner points



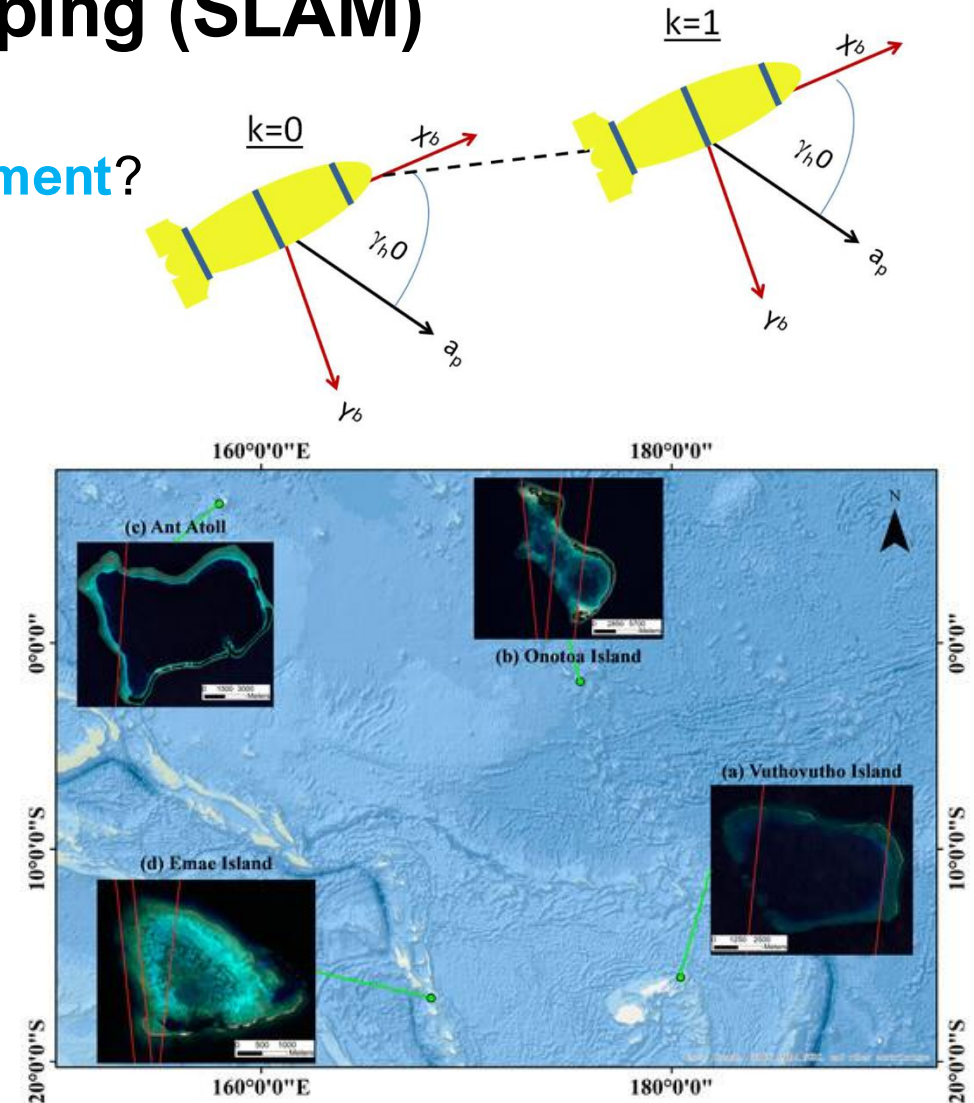
Extract 3D position of 16 corner points



Estimate the sonar camera extrinsic by solving PnP algorithm for 2D image pixel and 3D sonar point correspondence.

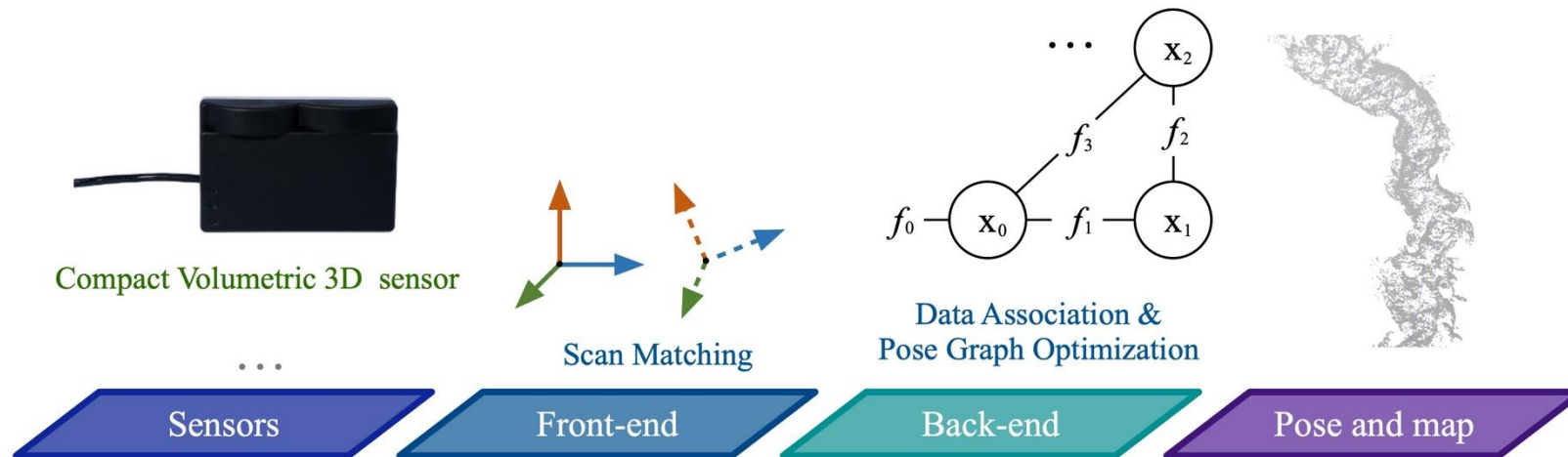
Simultaneous Localization and Mapping (SLAM)

- Why we need **SLAM** for the **underwater environment**?
 - GPS is unavailable underwater
 - Dead reckoning alone drifts quickly
- Why not rely only on bathymetric maps?
 - Bathymetric maps are too coarse
 - Coastal bathymetry changes over time
 - Bathymetry may includes repeated patterns



Simultaneous Localization and Mapping (SLAM)

- The robot builds a map of an **unknown** environment while at the same time estimating its own **position** within that **map**.
- We propose a **LiDAR inspired** 3D sonar SLAM pipeline :



Data association: how a robot decides whether a new observation corresponds to something it has already seen before or something new.

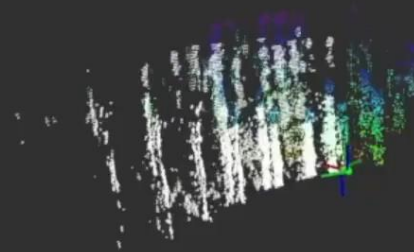
Pose graph optimization: how to adjust the robot's estimated positions so that all relative motion and observation constraints agree as well as possible.

Results

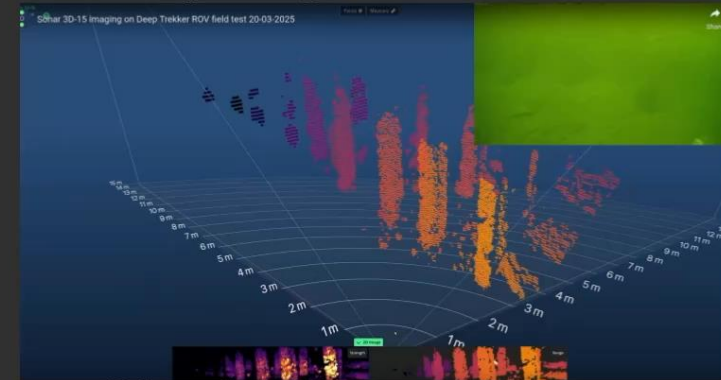
Sonar registration in pier piling environment



Pier piling environment



2x



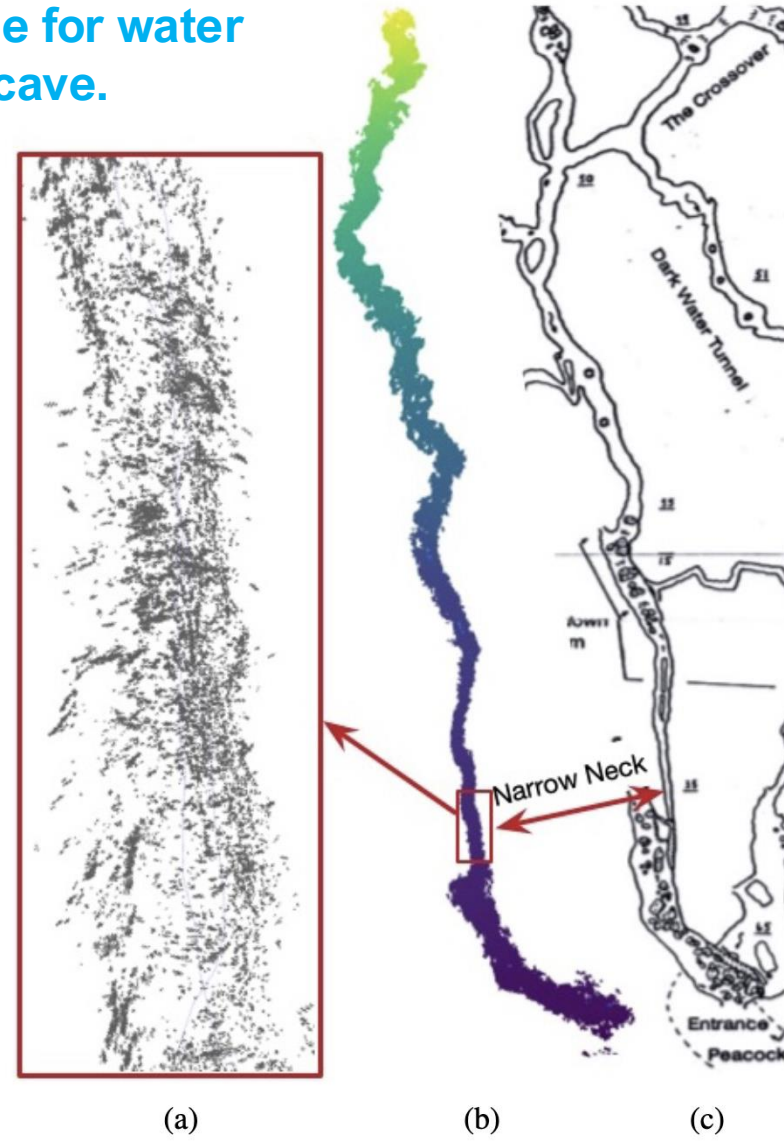
Single 3D sonar scan
& camera visibility

Results

Serves as a spatial reference for water quality mapping within the cave.



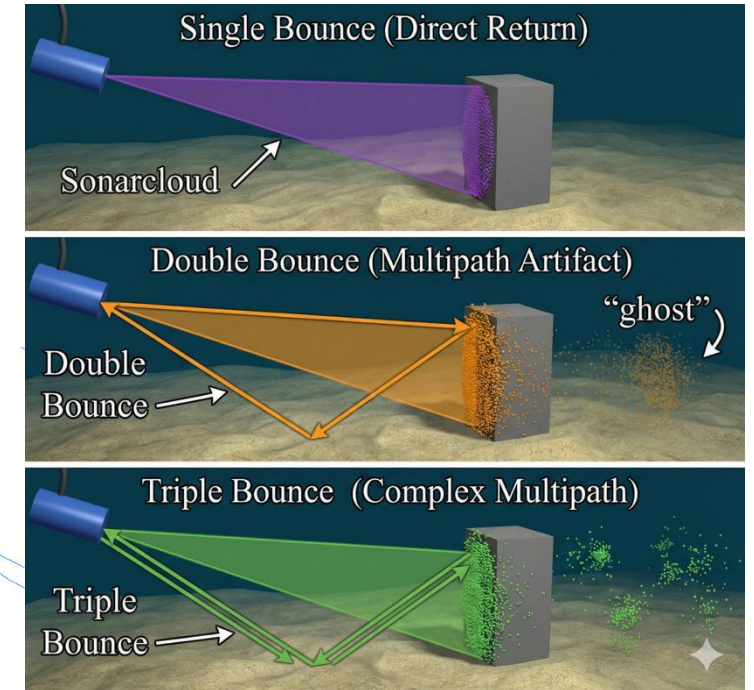
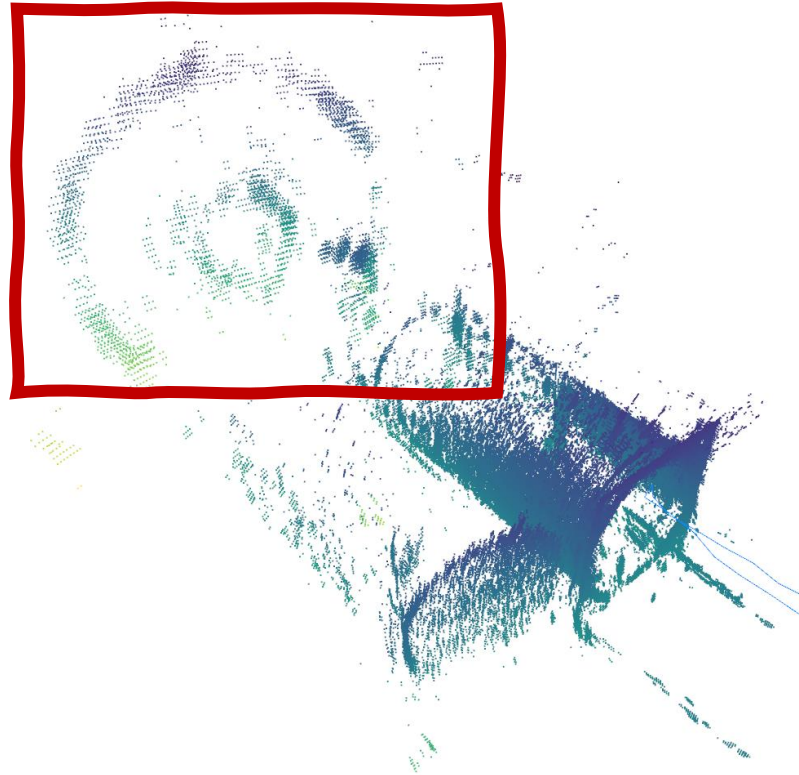
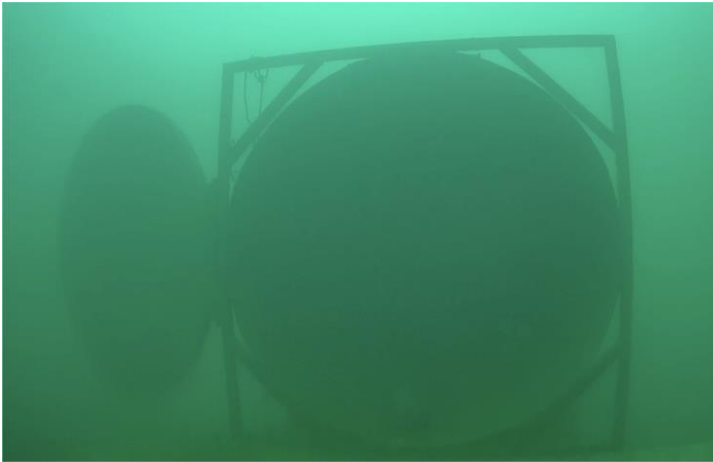
Ginnie Spring, FL



Peacock Cave, FL

Results

- **Multi-path effect:** signal distortion caused by reflections, where indirect signal paths interfere with the direct path, leading to localization, perception, or communication errors.



Additional challenge for multi-robot case

Underwater Unmanned Vehicles (UUV)

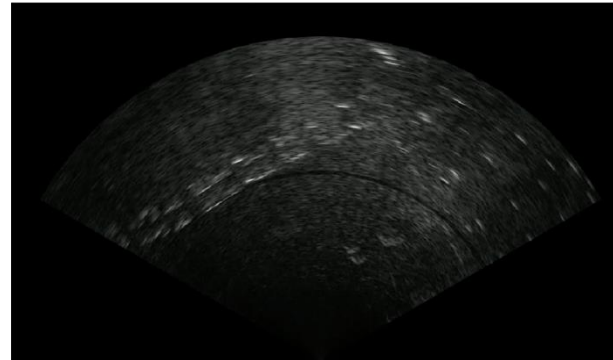
- **No** GPS (Seawater strongly attenuates radio waves)
- Influenced by **hydro-dynamics** (robot would rotate/drift)
- Only sonar in murky water
- **Unstructured** Features

Multi UUV

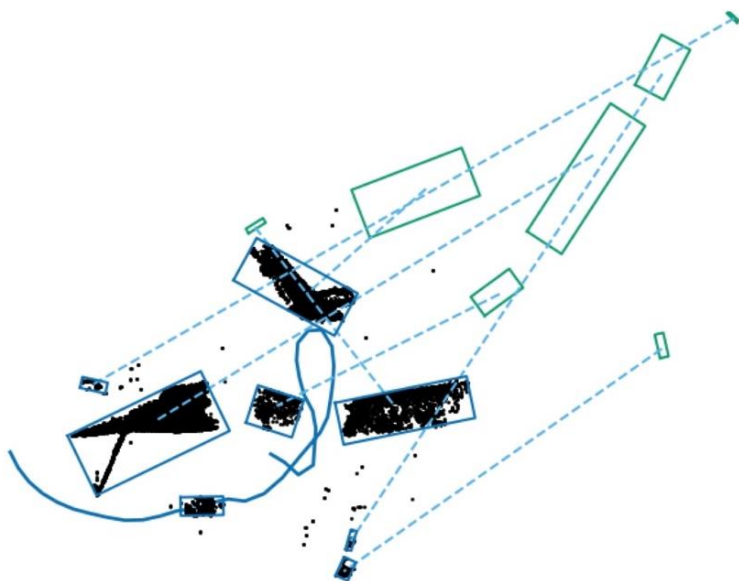
- Even high-speed(hs) acoustic modems are limited to **62.5 kbps** (Wi-Fi at 10–1000 Mbps).
- Rendezvous is not allowed due to **acoustic interference from identical-frequency sonars**



<https://www.evologics.com/acoustic-modems>
EvoLogics underwater acoustic modems.



My solution: Object Graph Matching



Vertex-wise Comparison $d_v(\cdot)$

Edge-wise Comparison $d_e(\cdot)$

Utility Matrix U_{2D}

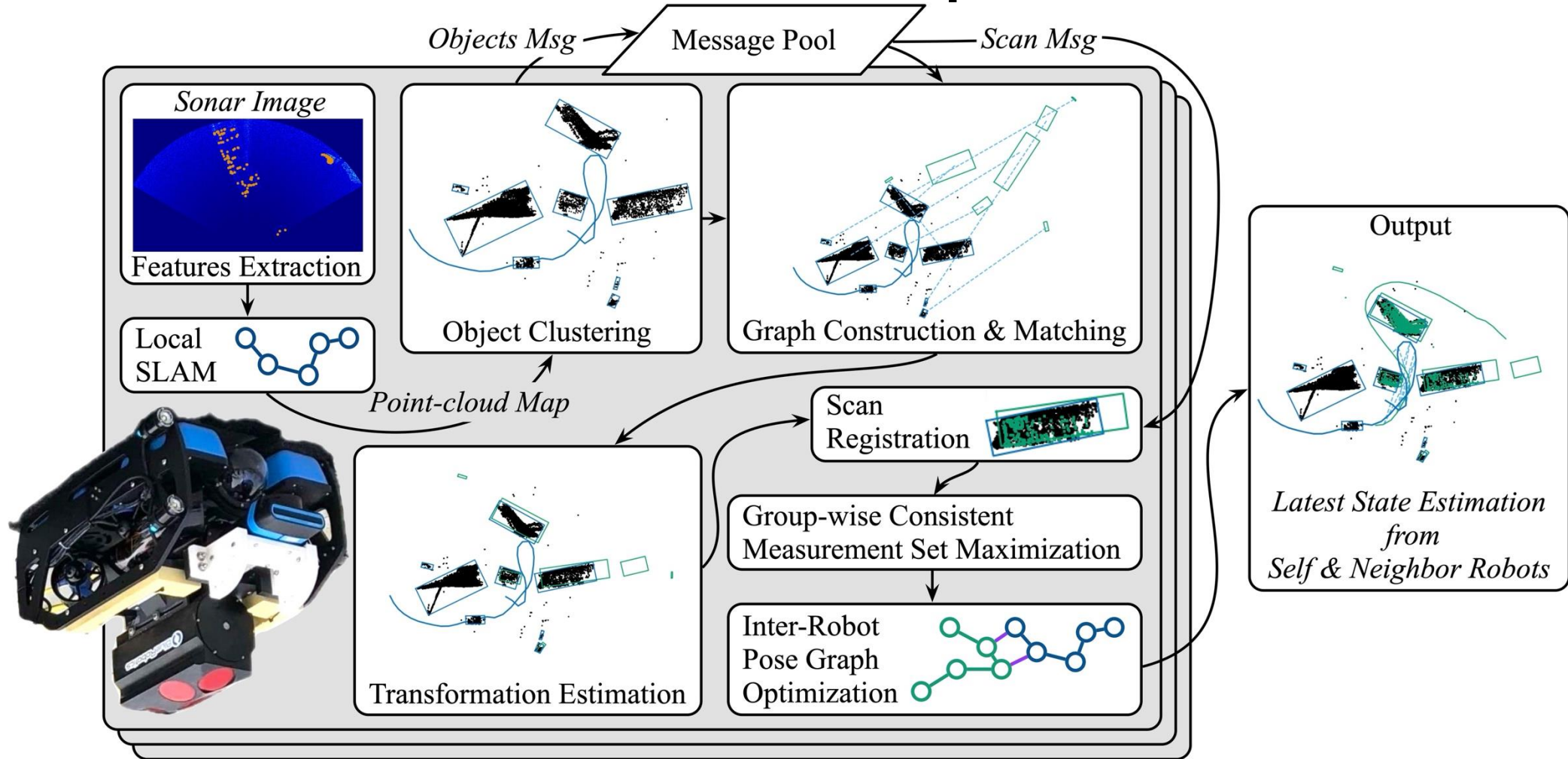
Reshape

Principal Eigenvector A_e^*

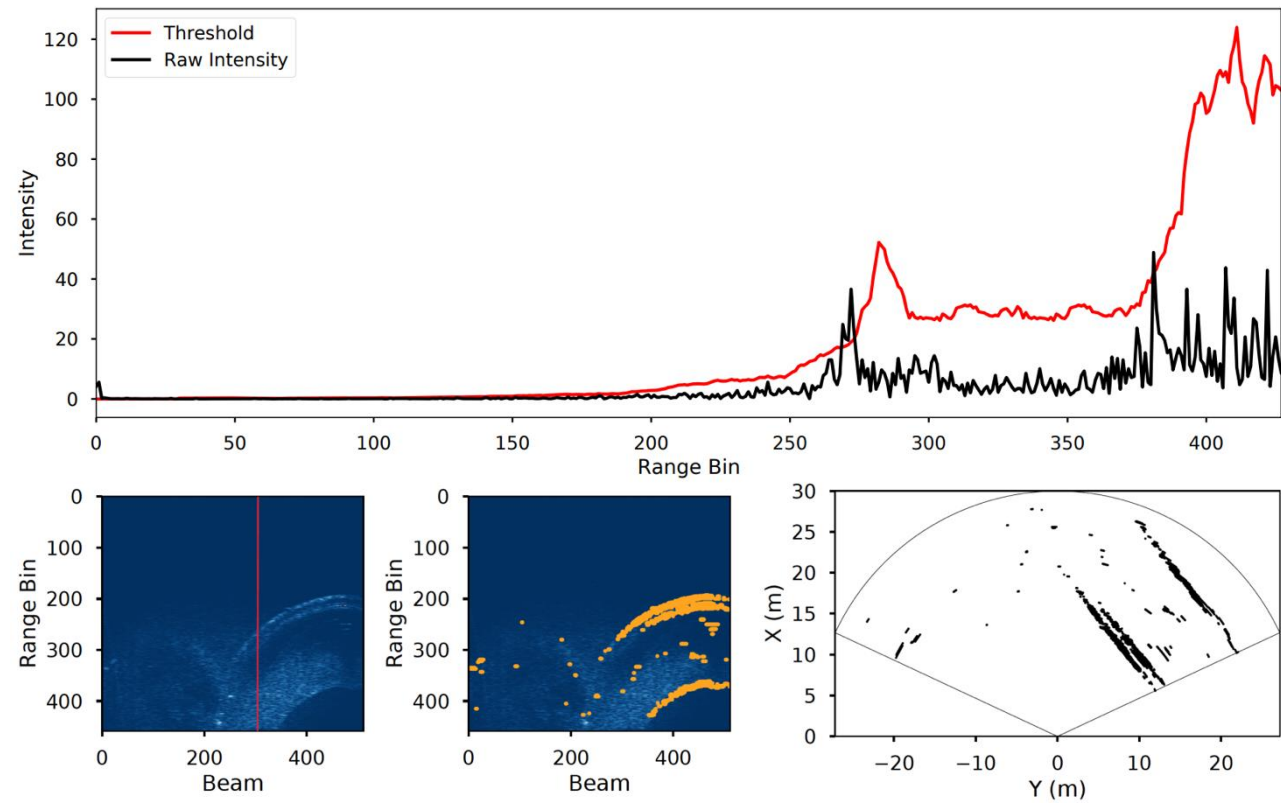
0	0	0.2	0	0	0.2	0.2	0.7	0	0	0	0.1
---	---	-----	---	---	-----	-----	-----	---	---	---	-----

Process of Object Graph Matching.

An Overview of the DRACo-SLAM2 Pipeline



Feature Extraction: SOCA-CFAR (Smallest Of Constant False Alarm Rate)



Feature Points in Polar Coordinate.

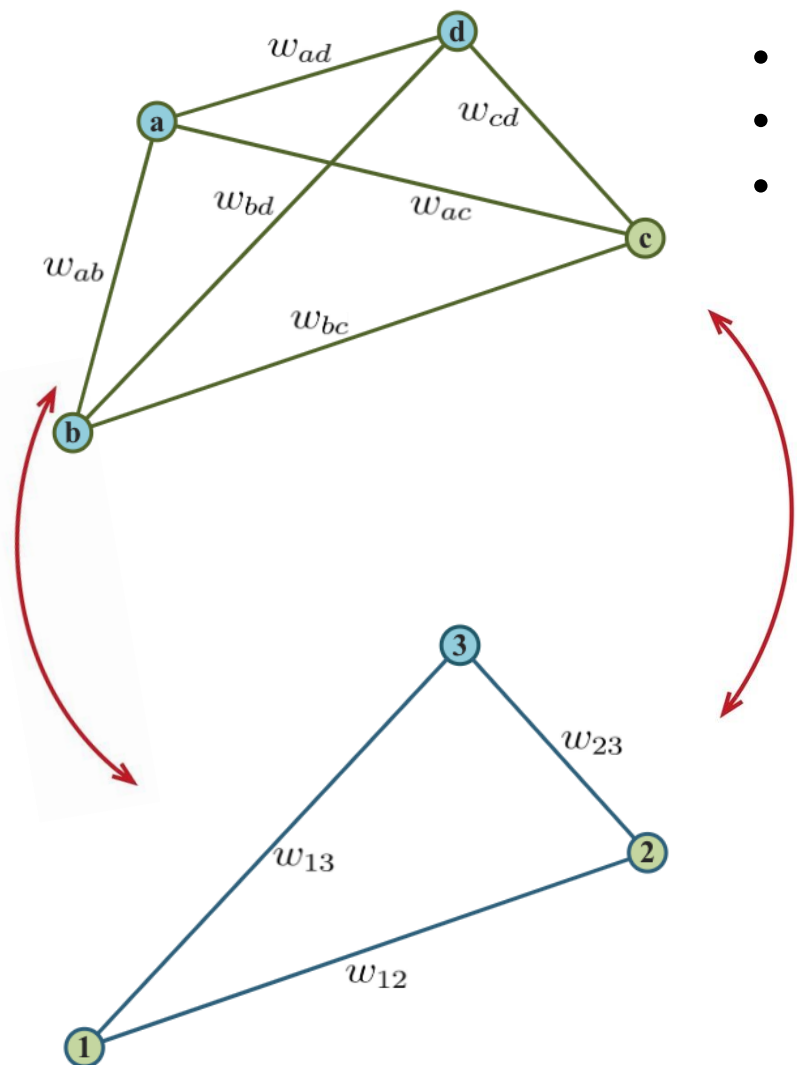
Feature Points in
Cartesian Coordinate.

Sonar image from an Oculus M750d with 512 beams.

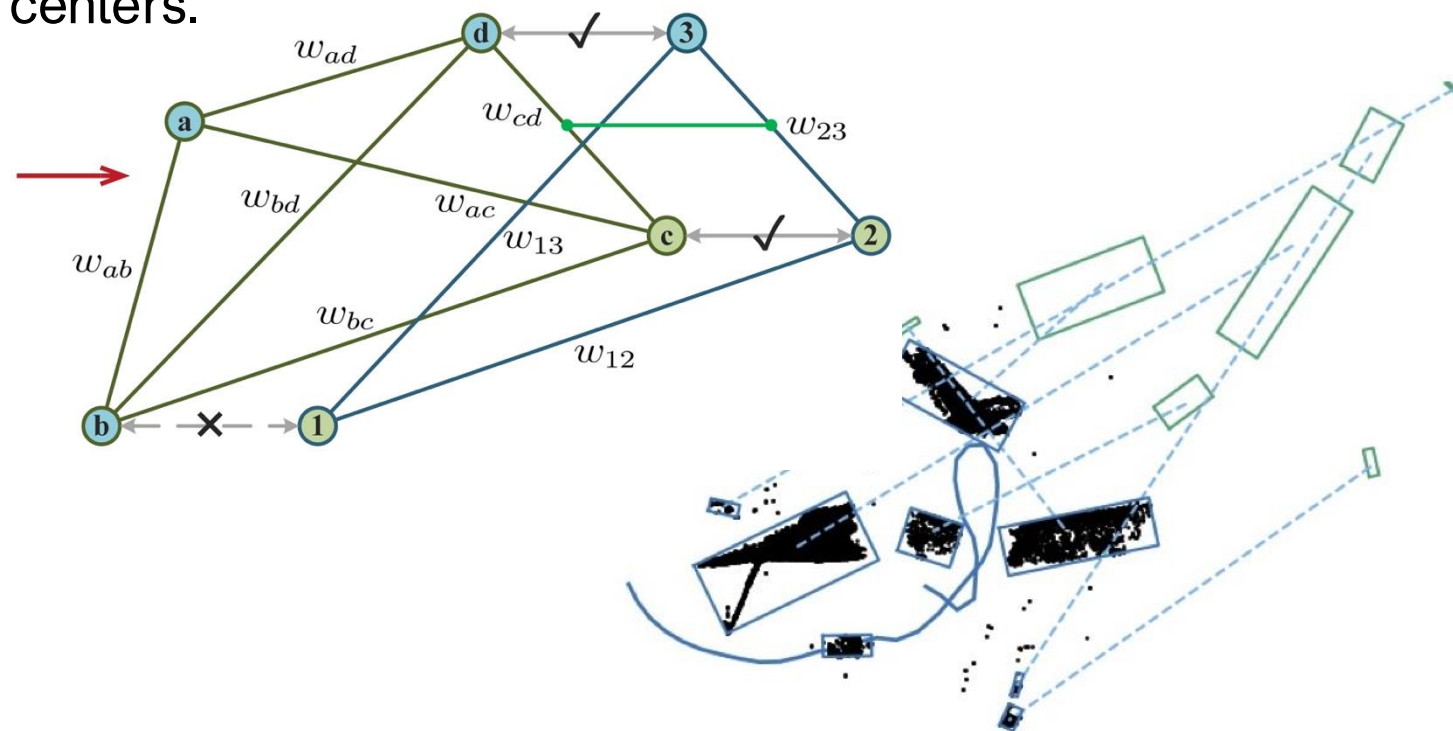


In satellite images, different surfaces
interact with light differently.

Inter-Robot Object Matching

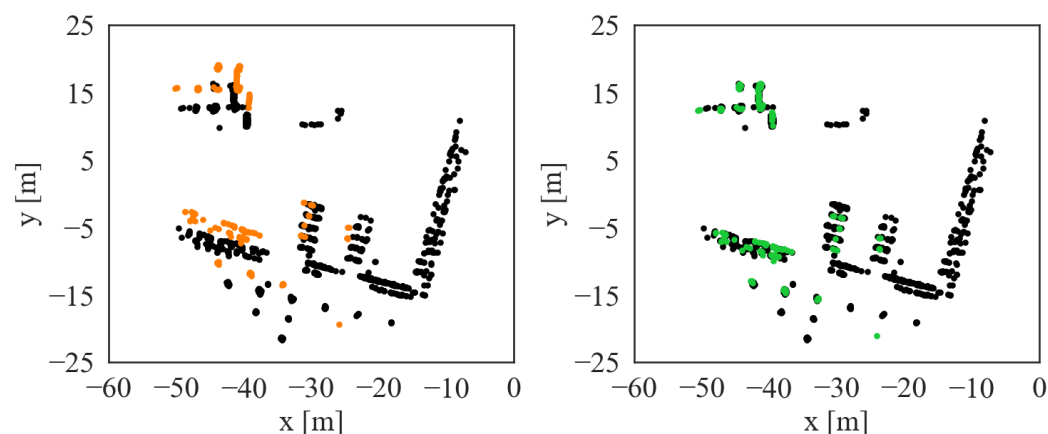


- DBScan Algorithm for object clustering
- Vertex-wise Comparison: Bounding box size;
- Edge-wise Comparison: Relative distance between object centers.



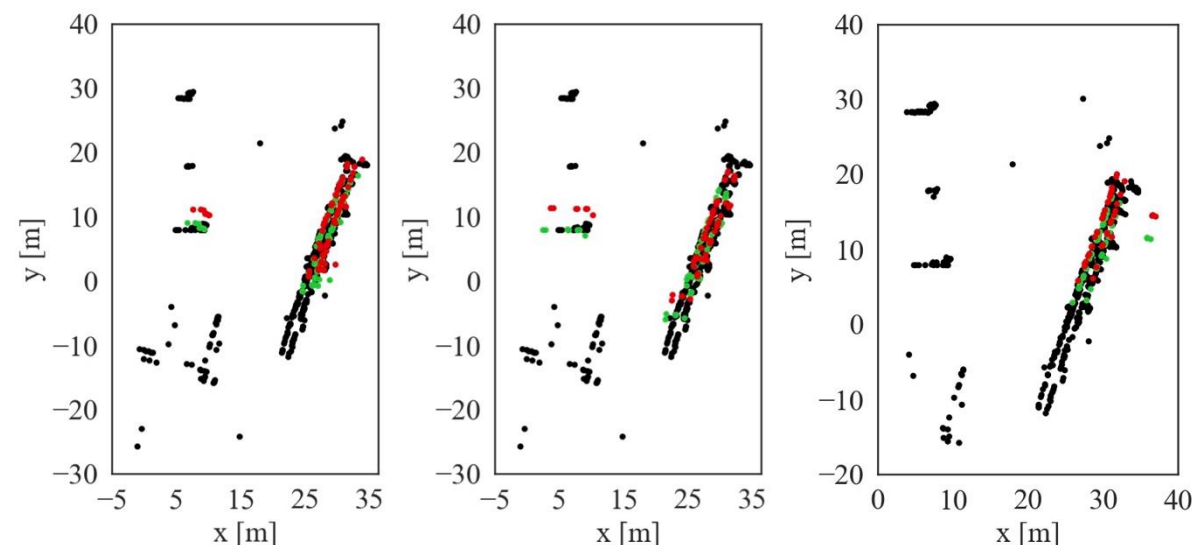
Matched results with local robot objects shown in blue and neighbor robot objects in green.

Outlier Check: GCM



• Source Cloud • Target Cloud • Source Cloud (Registered)

Success: The source cloud (orange) is roughly transformed into the local robot coordinate system using the transformation estimated from object graph matching.

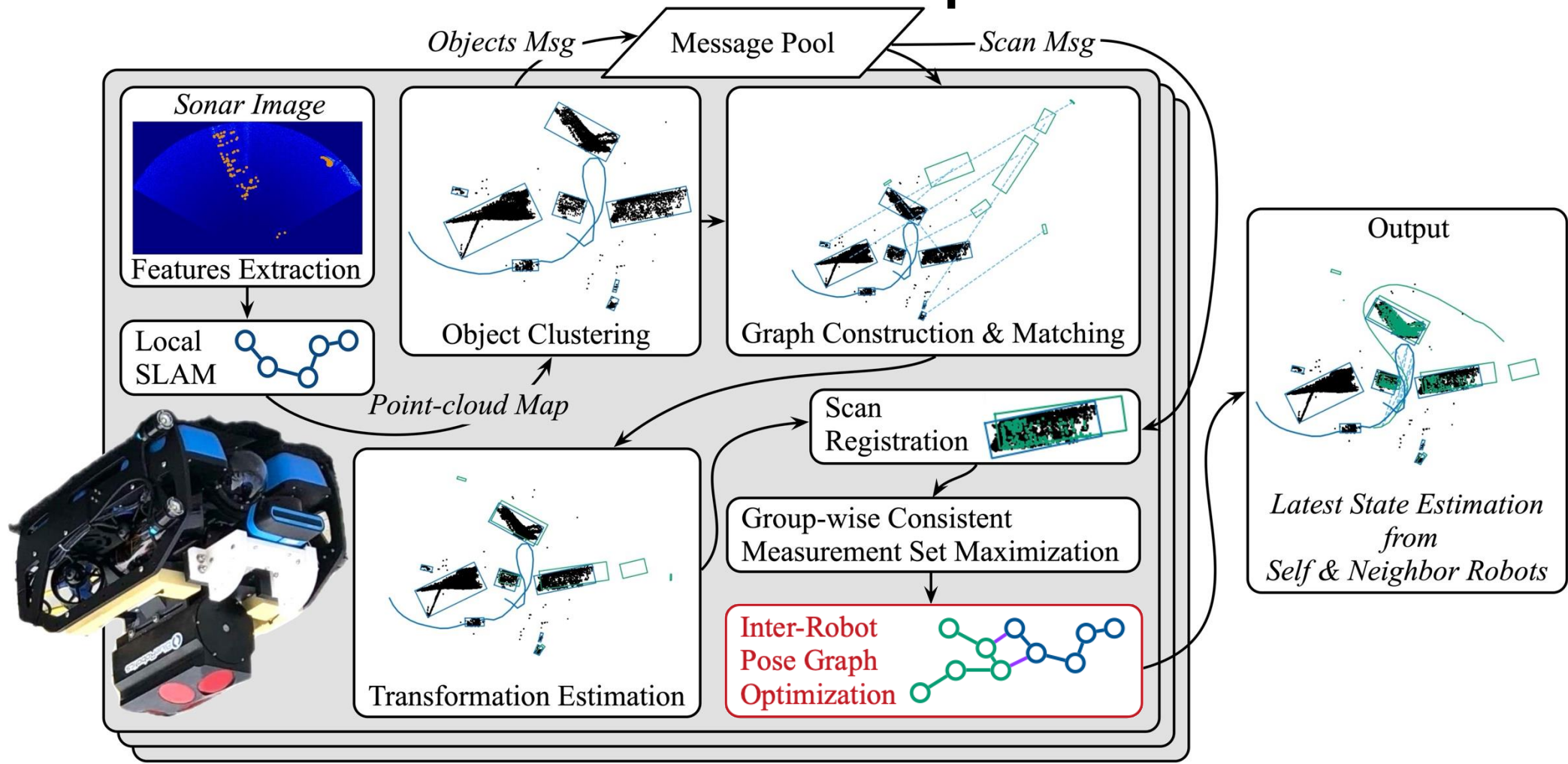


• Target Cloud • Source Cloud (Registered) • Source Cloud (GT)

Failure: A series of three inter-robot loop closures, in the same region of the environment are impacted by **similar point cloud registration errors**.

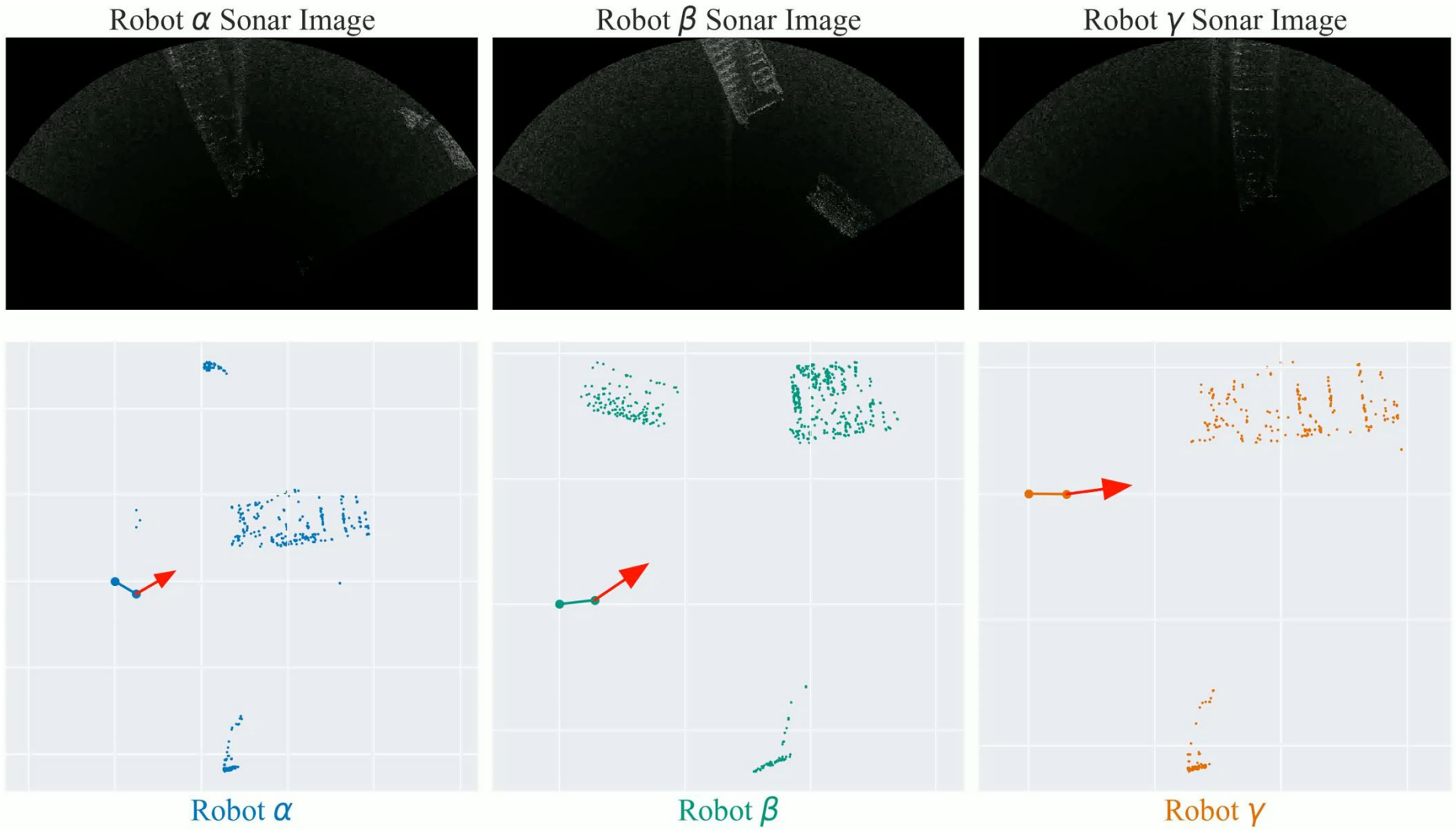
GCM is used to detect mis-associated loop closures.

An Overview of the DRACo-SLAM2 Pipeline



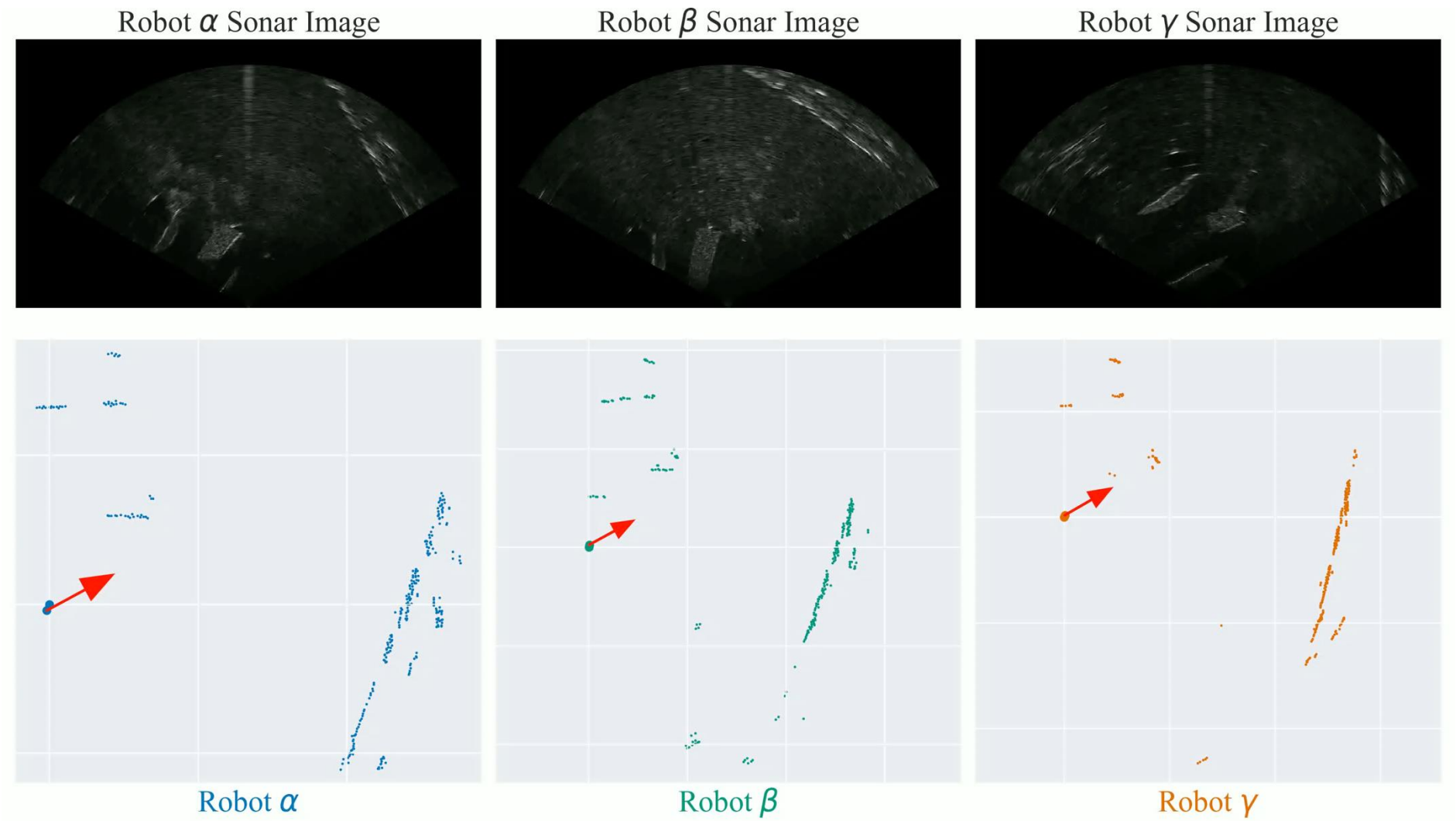
The loop closures accepted by GCM are added into the pose graph optimization.

Performance on Airplane Simulation Dataset



Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.

Performance on USMMA Real Dataset



Shown above are all sonar image keyframes from this dataset, and the corresponding SLAM results.

Autonomous Exploration of ASV

- Characteristics:
 - GPS available but **degraded**
 - **Drift** from currents and wind
 - Vision and LiDAR fusion
 - **Uneven** structures
- Core environmental sensors for ASVs
 - **Water quality:** Conductivity, Temperature, Depth
 - **Acoustics:** bathymetry and currents
 - **Optical sensing:** surface imagery and thermal data
 - **Meteorology:** air–sea interface measurements
- LiDAR-based algorithms can provide reliable localization and mapping in structured environments but tend to drift in structureless environments.
- **How to explore smartly with limited energy and computation resources?**

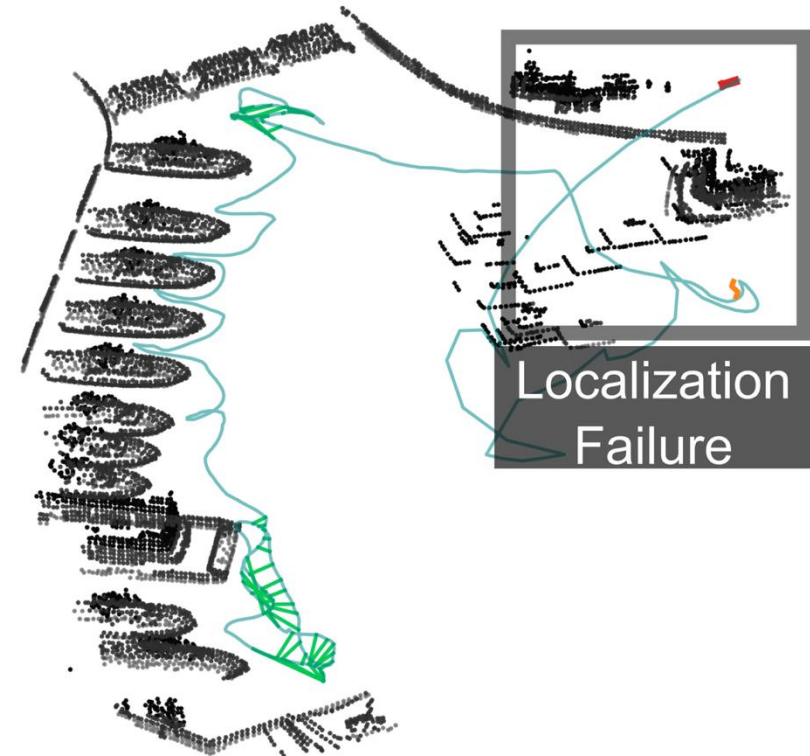


How to explore smartly with limited energy and computation resources?

Exploration: The robot chooses its own path to discover new areas while staying safe and localized.

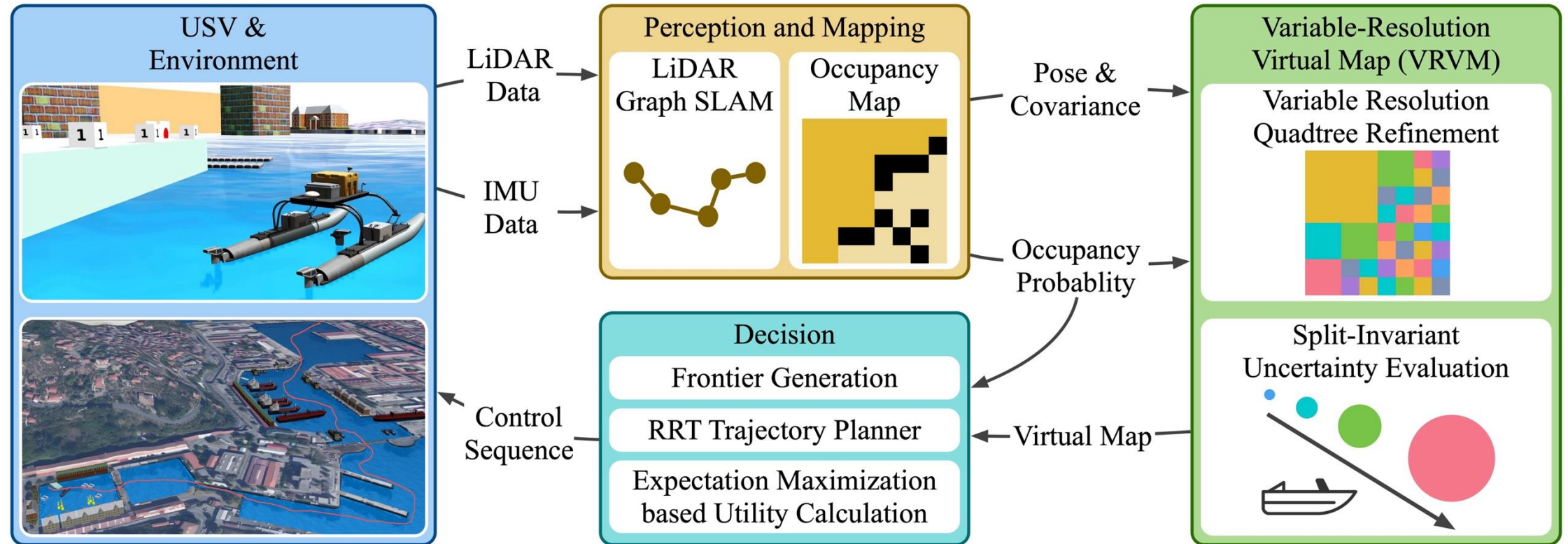
How to quantify “exploring smartly”?

- Designed for onboard **embedded** deployment
- Robust and consistent **localization** (where it is) and **mapping** (what the environment looks like)
- Finish the task as **fast** as possible

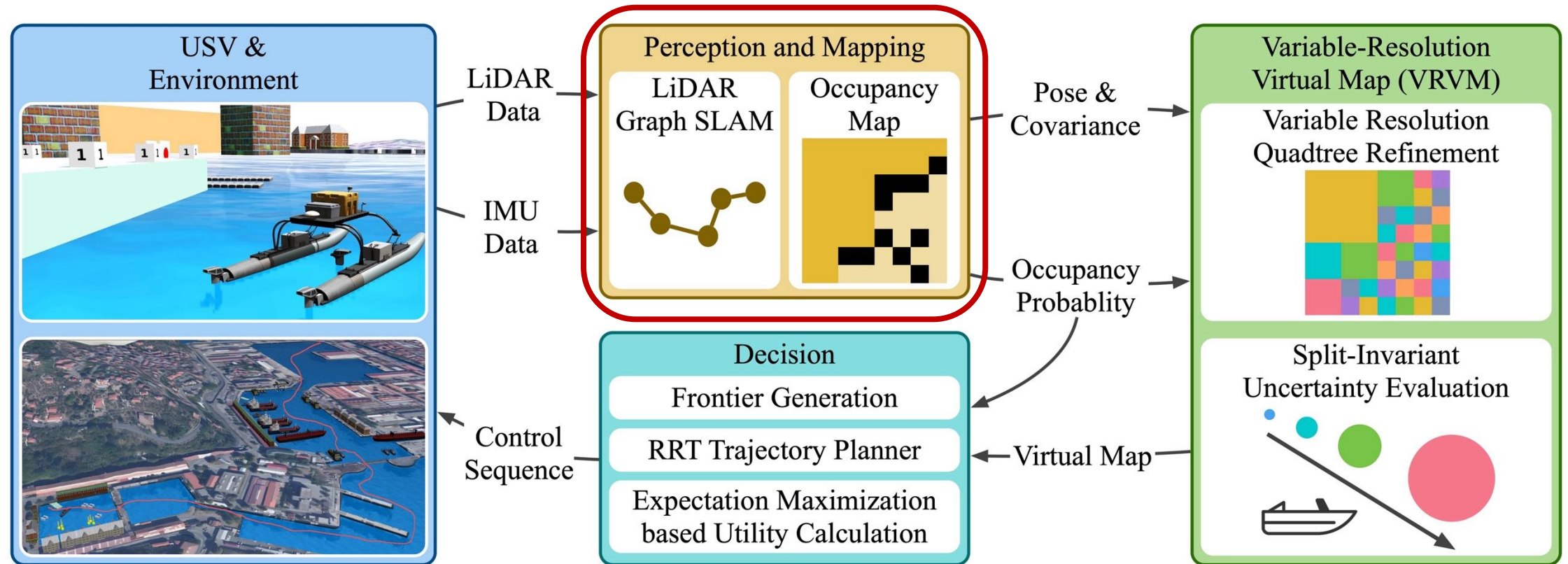


Drift from the SLAM localization failure

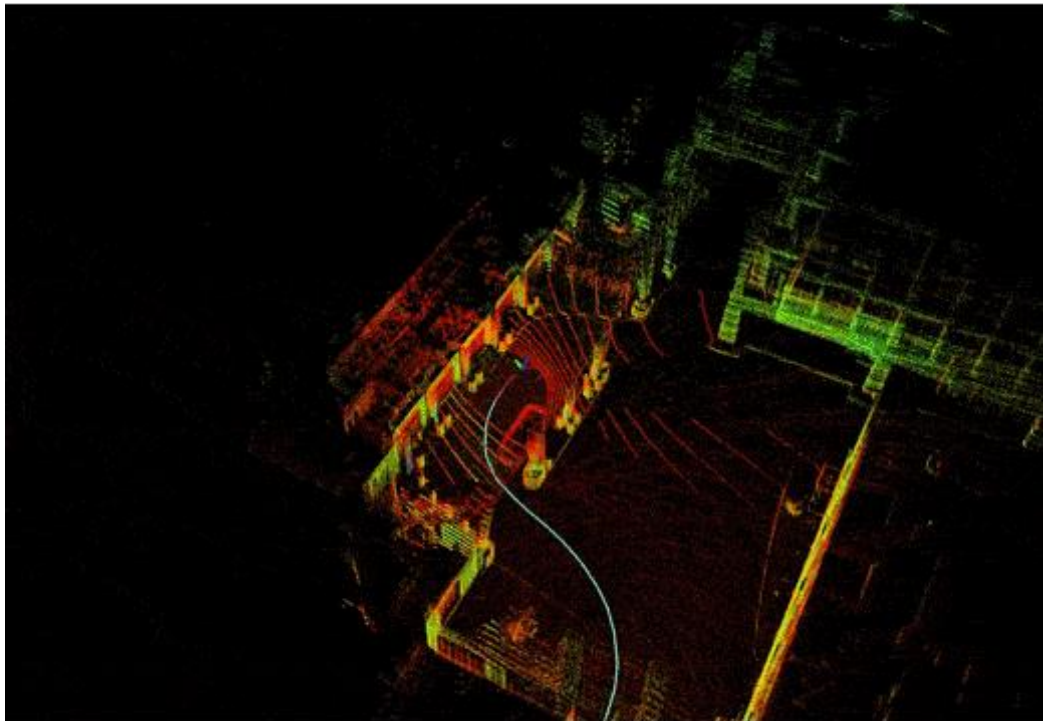
Pipeline for Variable Resolution Virtual Map



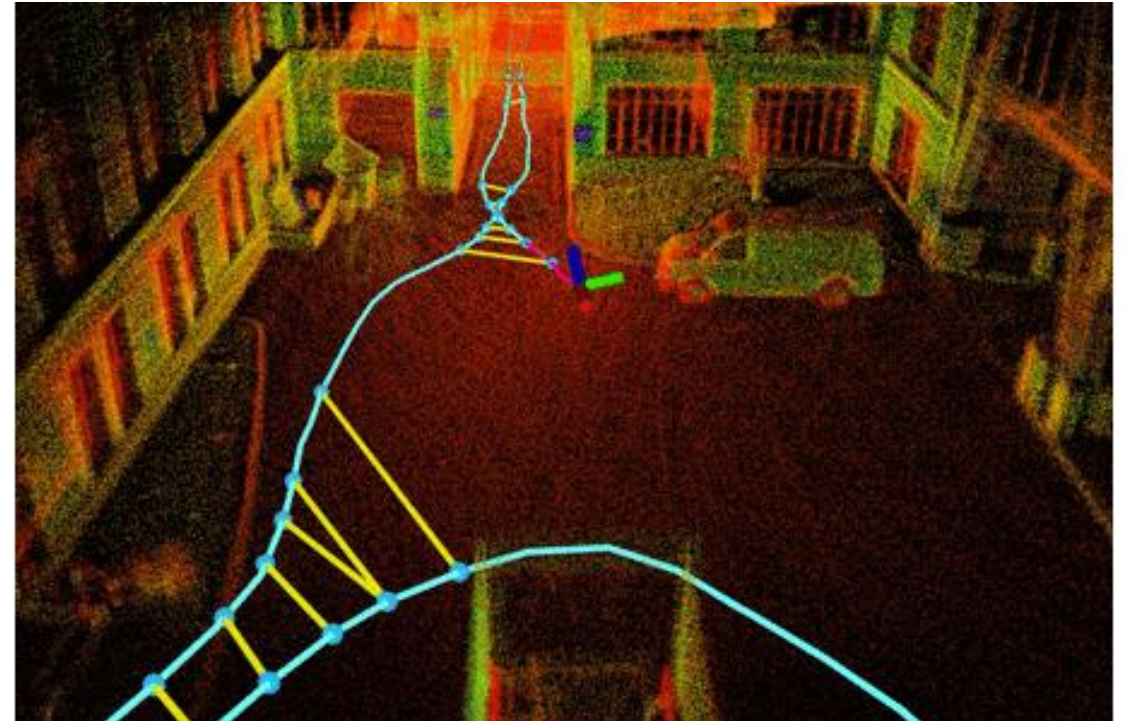
Pipeline for Variable Resolution Virtual Map



LiDAR Perception and Mapping



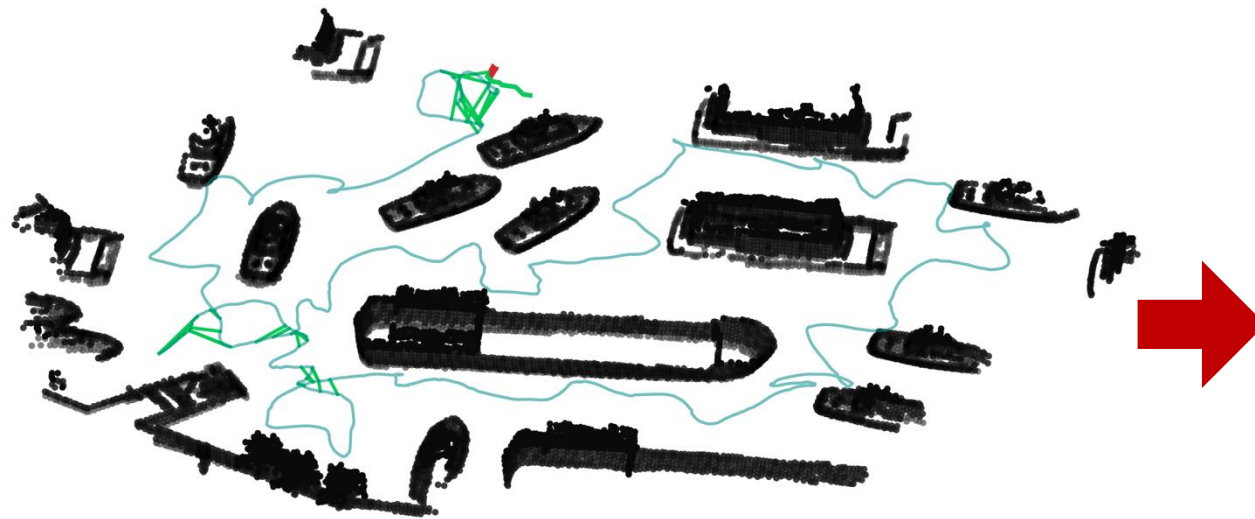
LiDAR Scan Registration



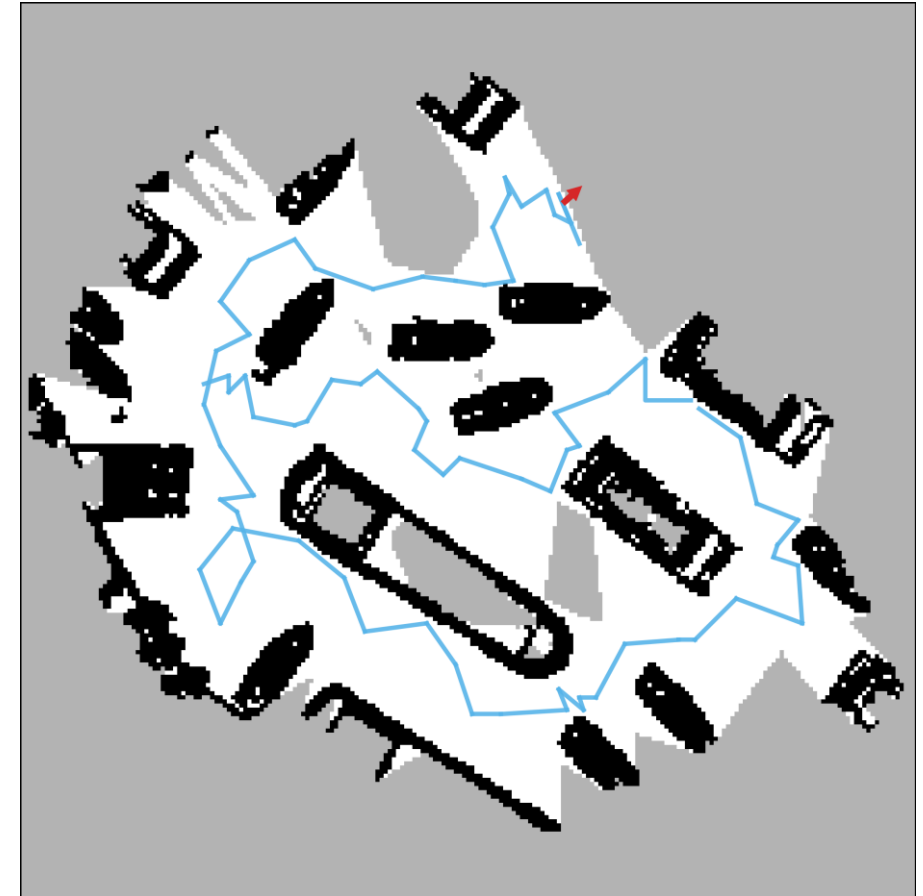
LiDAR Loop Closure &
Pose Graph Optimization (PGO)

Occupancy Mapping

- **Occupancy Map:** a map that indicates which areas are **free, occupied, or unknown**, based on sensor observations.

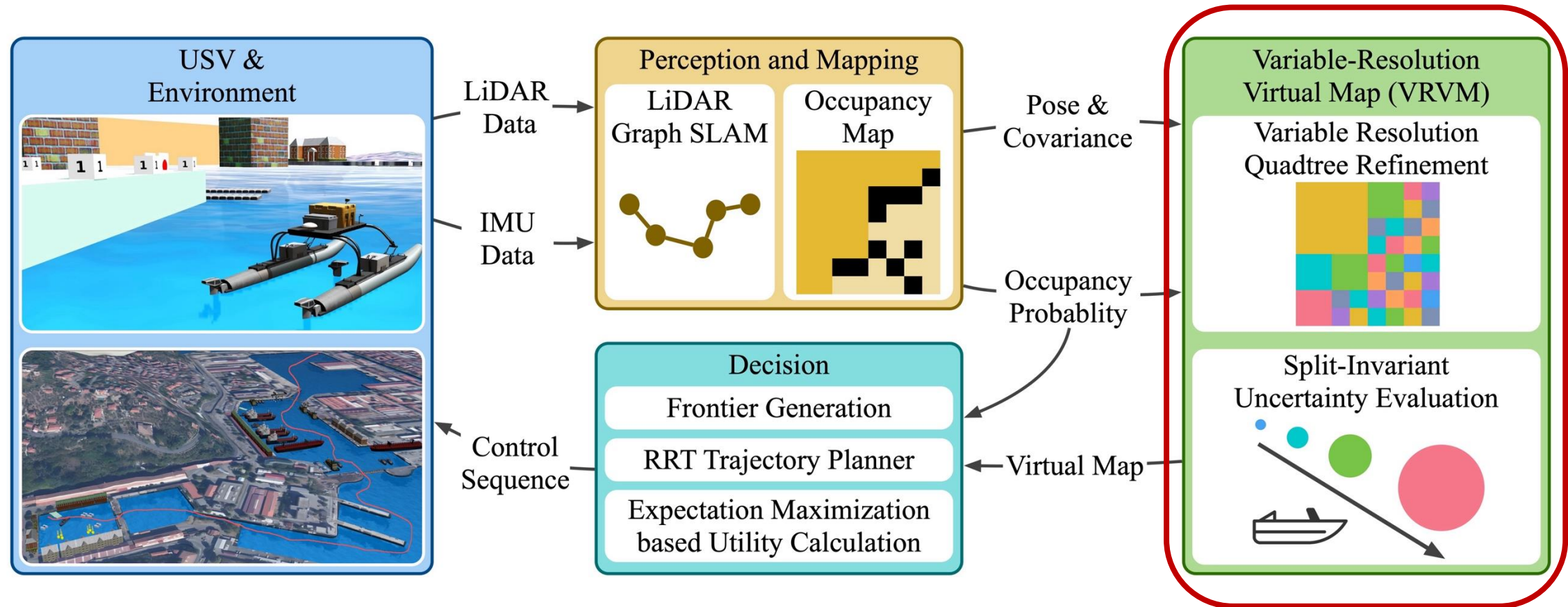


Localization and mapping result



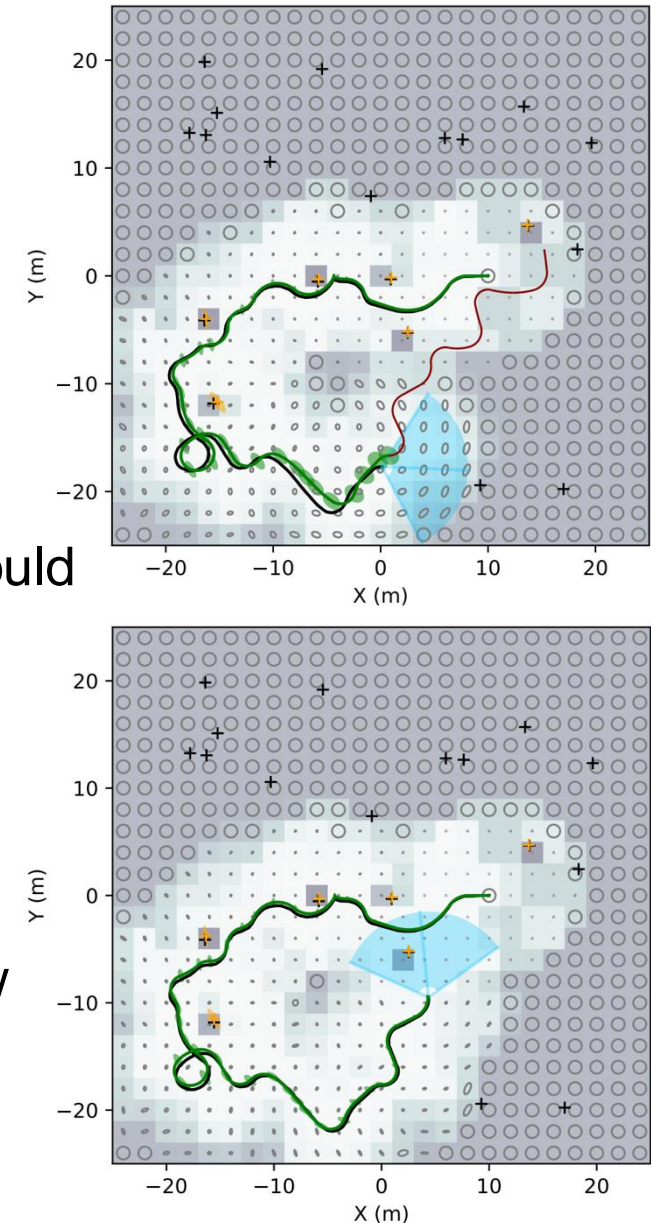
Corresponding occupancy map

Pipeline for Variable Resolution Virtual Map



Uniform Virtual Map based Exploration

- The robot's uncertainty about **localization** and **mapping** is represented by **covariance matrix** (a statistical measure of confidence) associated with its estimated position and map.
- A candidate goal is evaluated by predicting how uncertainty would grow along the way using **belief propagation**.
- This uncertainty is then used, together with travel distance, to decide whether the robot should **exploit** (return to previously visited areas to improve confidence) or **explore** (move into new areas to gather new information).



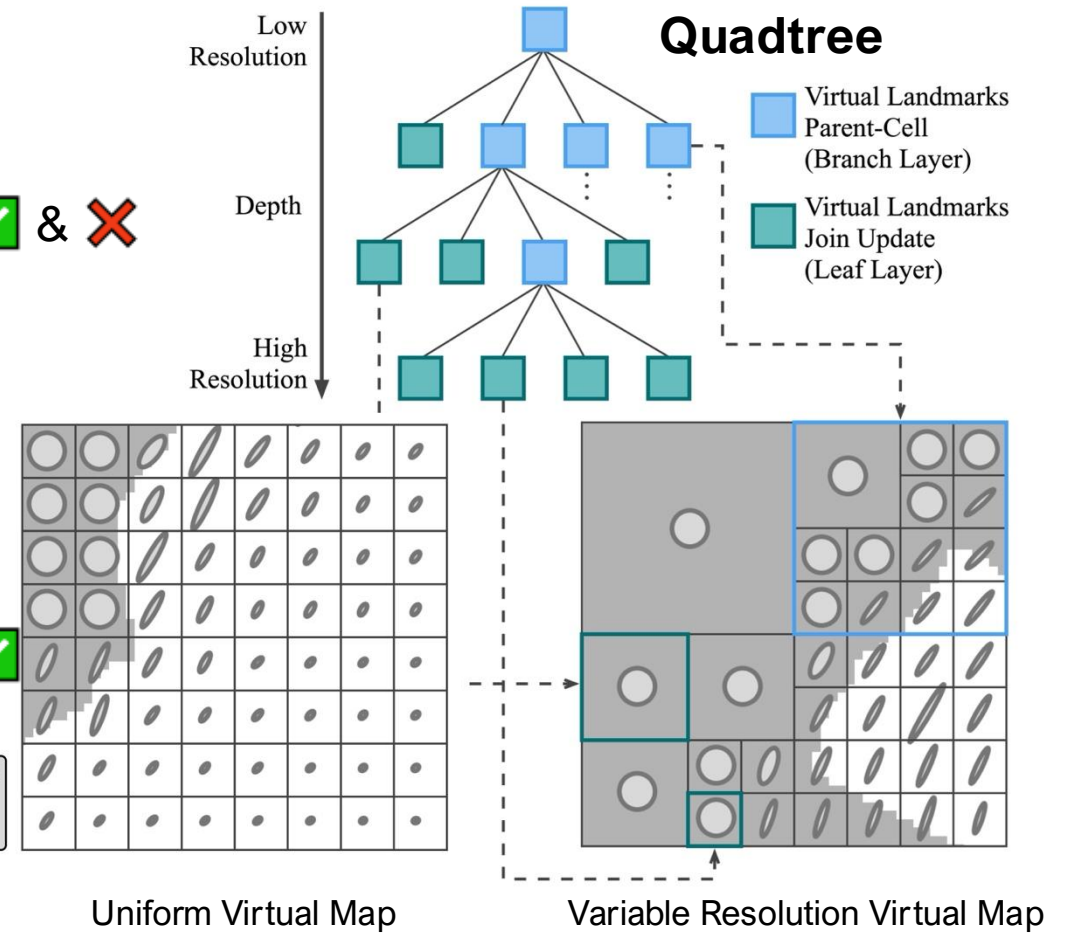
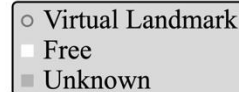
Variable Resolution Virtual Map

Uniform Virtual Map (UVM)

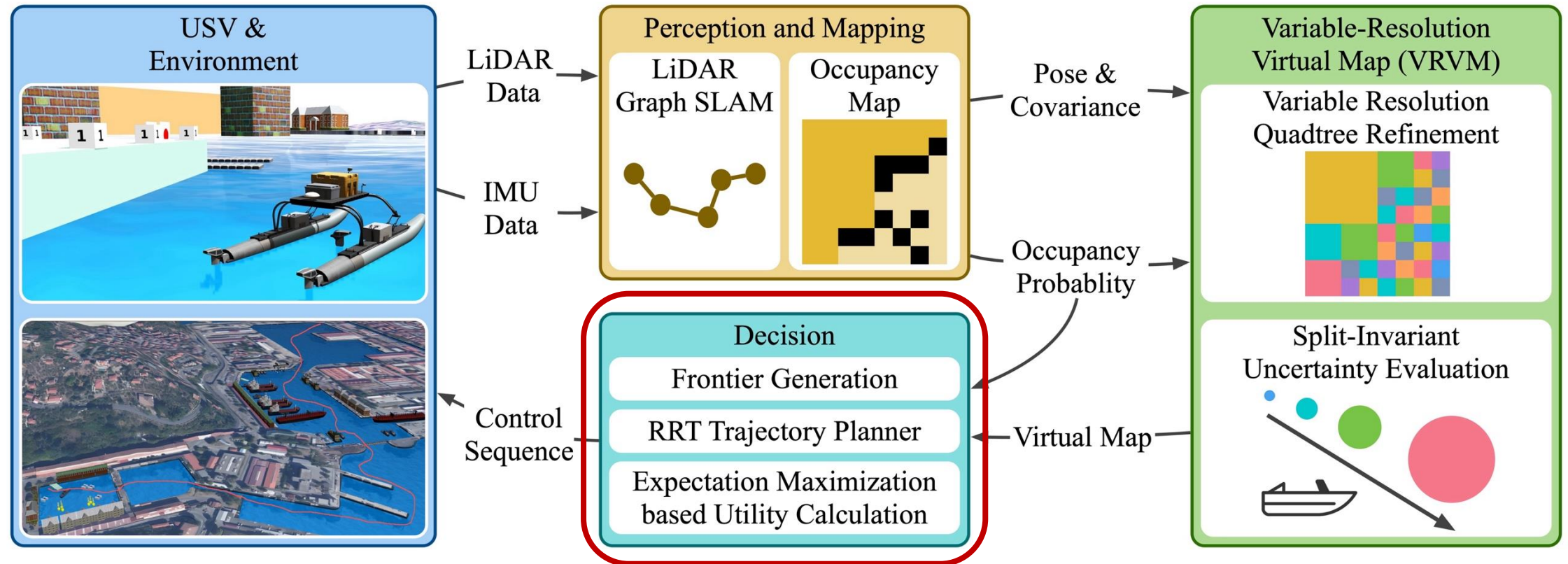
- Designed for onboard **embedded** deployment ❌
- Robust and consistent **localization** and **mapping** ✅ & ❌
- Finish the task as **fast** as possible ✅

Variable Resolution Virtual Map (VRVM)

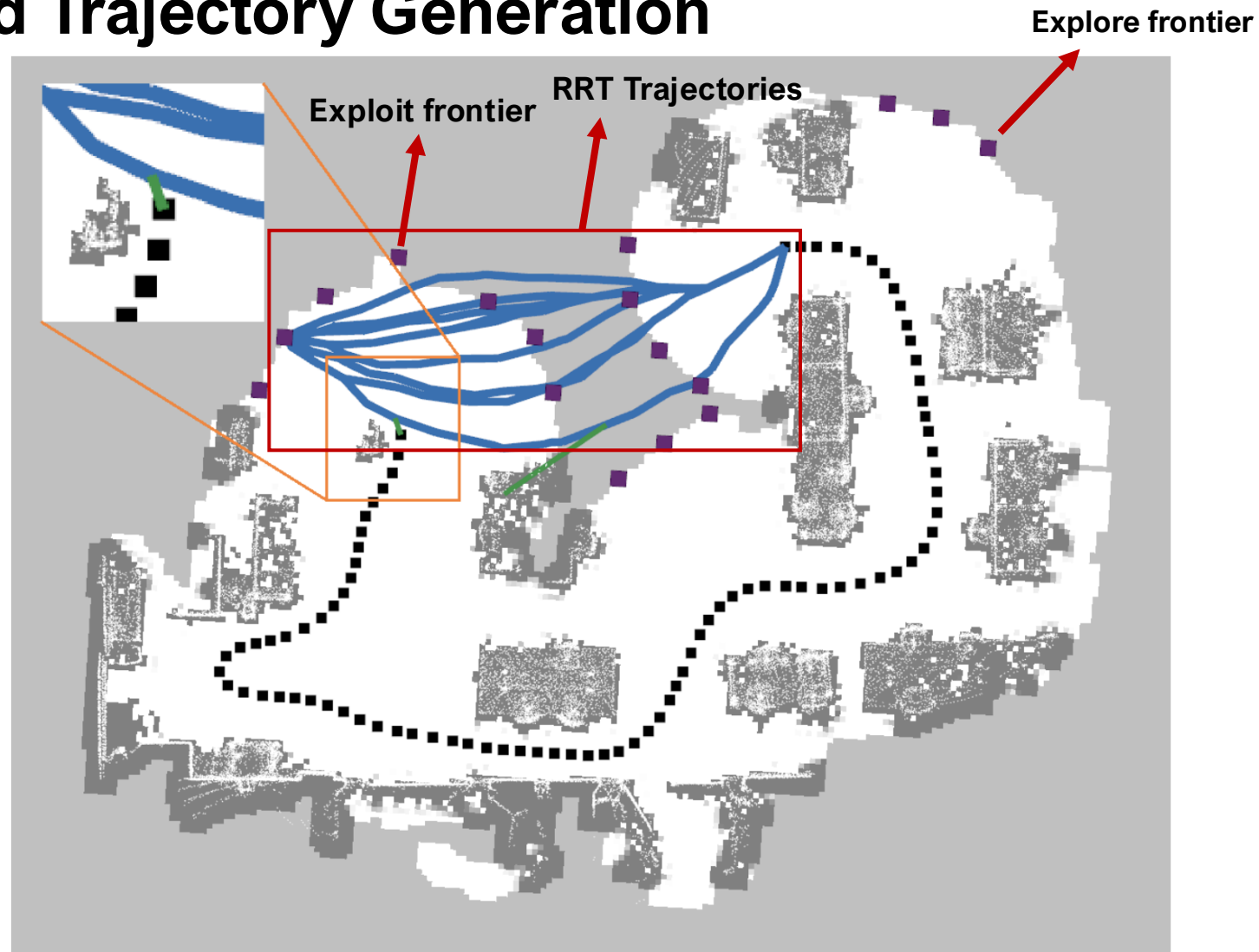
- Designed for onboard **embedded** deployment ✅
- Robust and consistent **localization** and **mapping** ✅
- Finish the task as **fast** as possible ✅



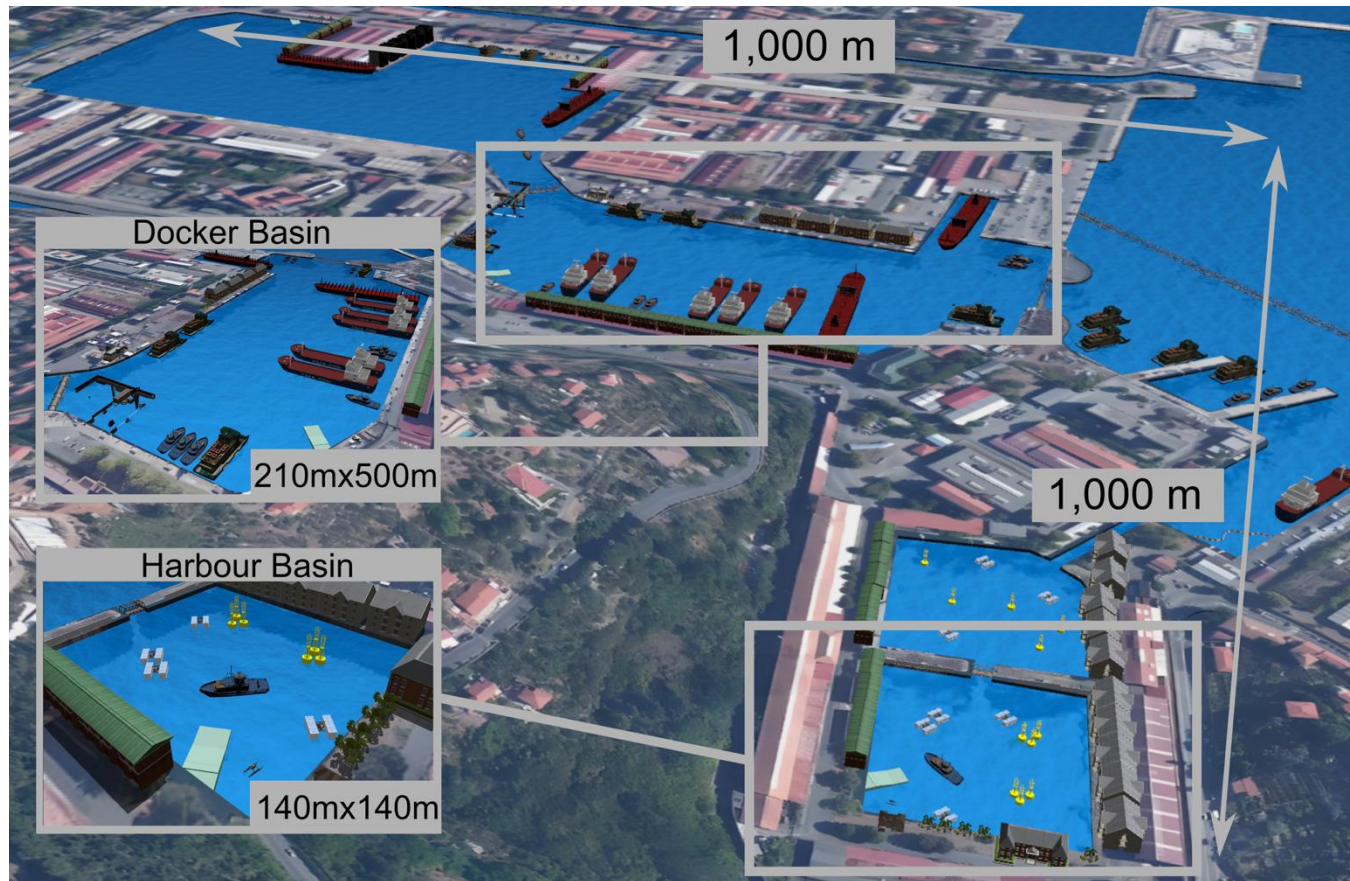
Pipeline for Variable Resolution Virtual Map



Frontier and Trajectory Generation



Performance in the VRX Simulation



Utility:

Selected Best Trajectory

$$\pi^* = \arg \max_{\pi \in \Pi} U(\pi),$$

$$U(\pi) = (U_{\text{map}}(\pi) + U_{\text{traj}}(\pi) + U_{\text{length}}(\pi)).$$

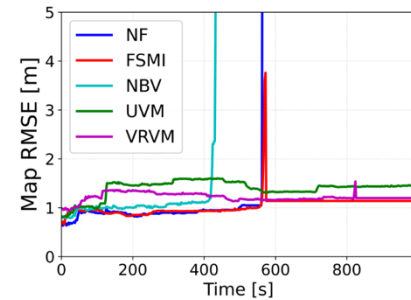
Map Uncertainty

Trajectory Uncertainty

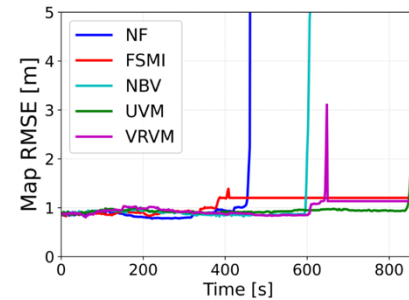
Travel Distance



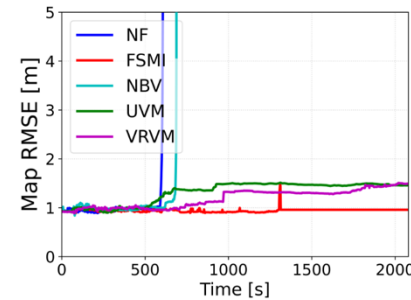
Performance in the VRX Simulation



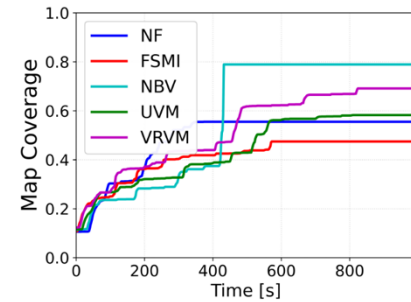
(a) Map Error - Sparse



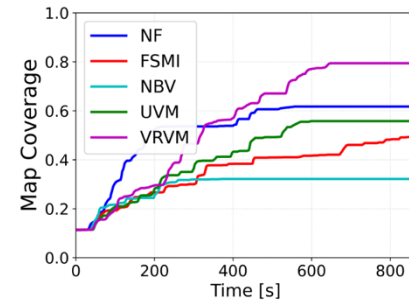
(b) Map Error - Moderate



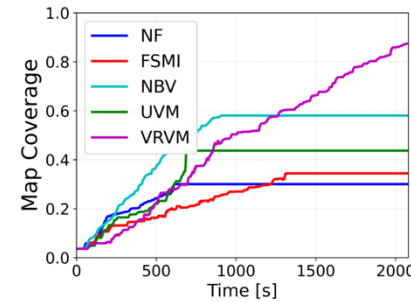
(c) Map Error - Dense



(d) Map Coverage - Sparse

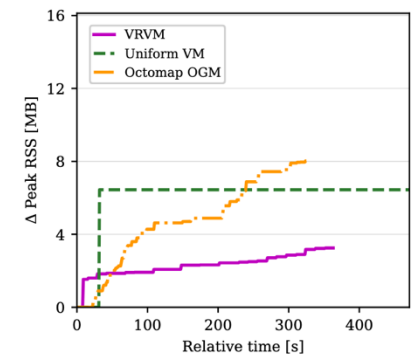
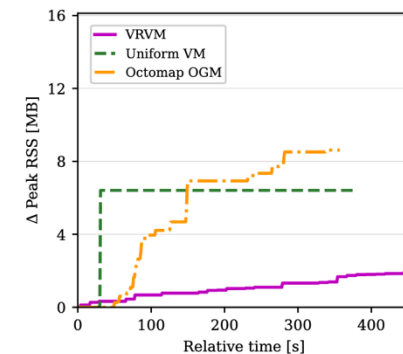
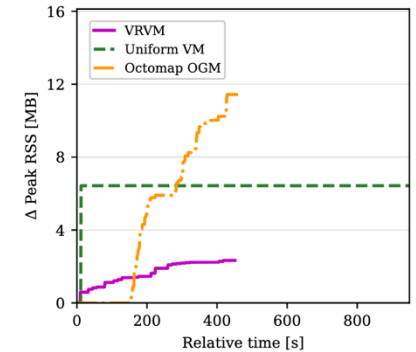
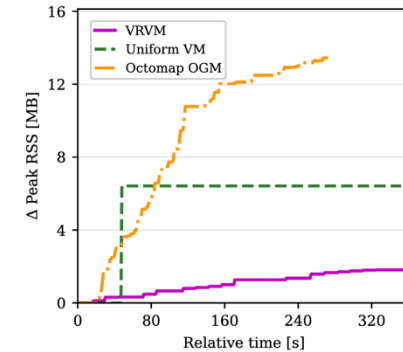


(e) Map Coverage - Moderate



(f) Map Coverage - Dense

Comparison of map efficiency and accuracy

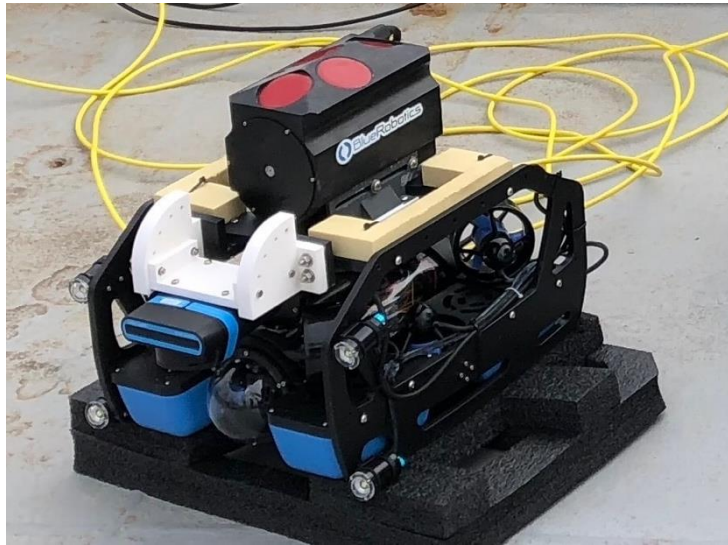


Comparison of memory usage

Ongoing and Next Step

- **UUVs with flexible grippers:** in situ environmental sampling
- **Multi-ASV exploration:** harbor security and surveillance
- **Large-scale ASV-UUV mission planning:** long-term environmental monitoring





Thank you!

